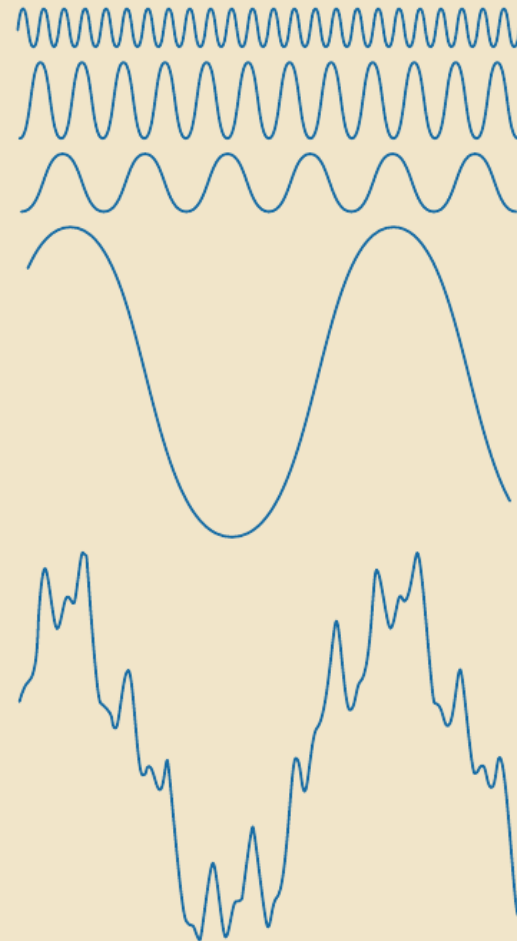


频域滤波

FIGURE 4.1

The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.



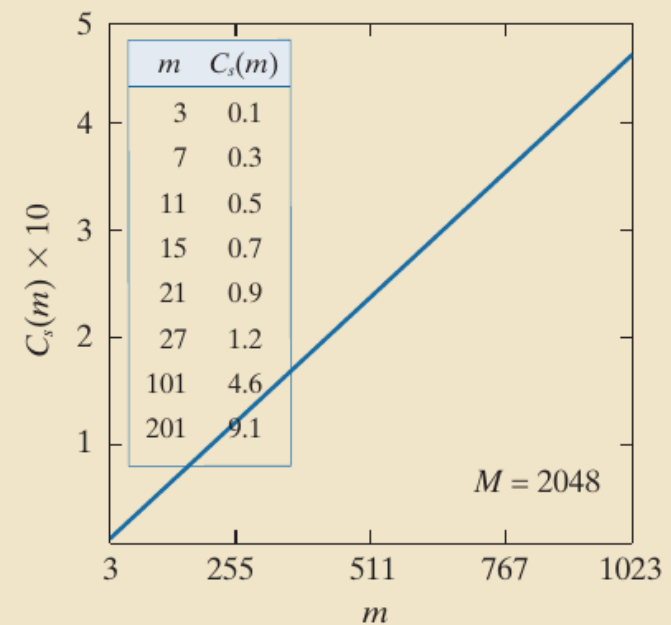
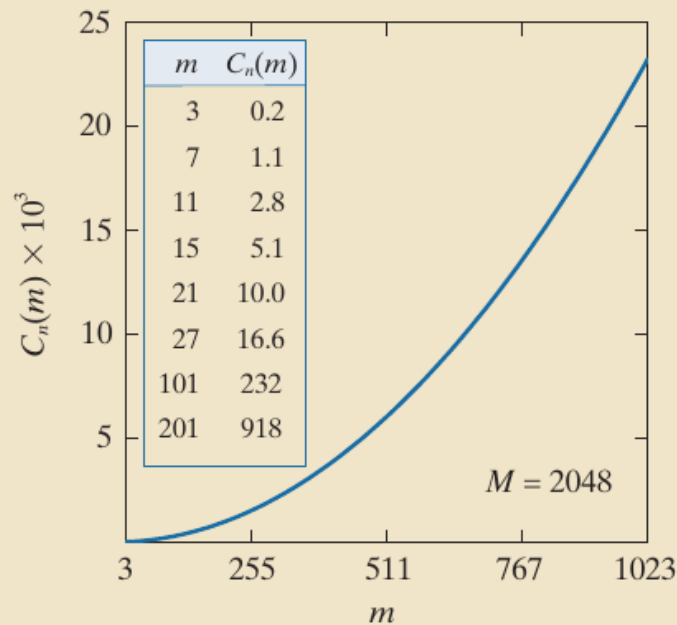
a b

FIGURE 4.2

(a) Computational advantage of the FFT over non-separable spatial kernels.

(b) Advantage over separable kernels.

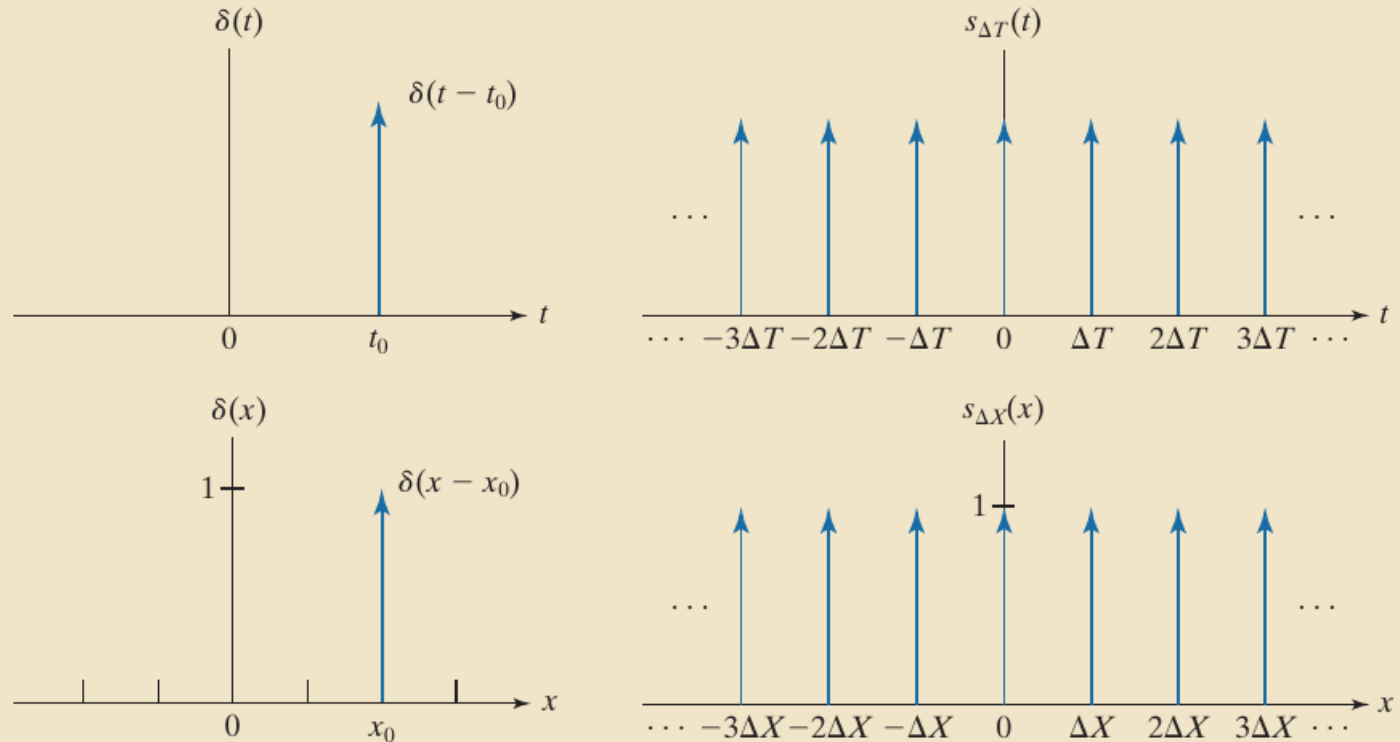
The numbers for $C(m)$ in the inset tables are not to be multiplied by the factors of 10 shown for the curves.

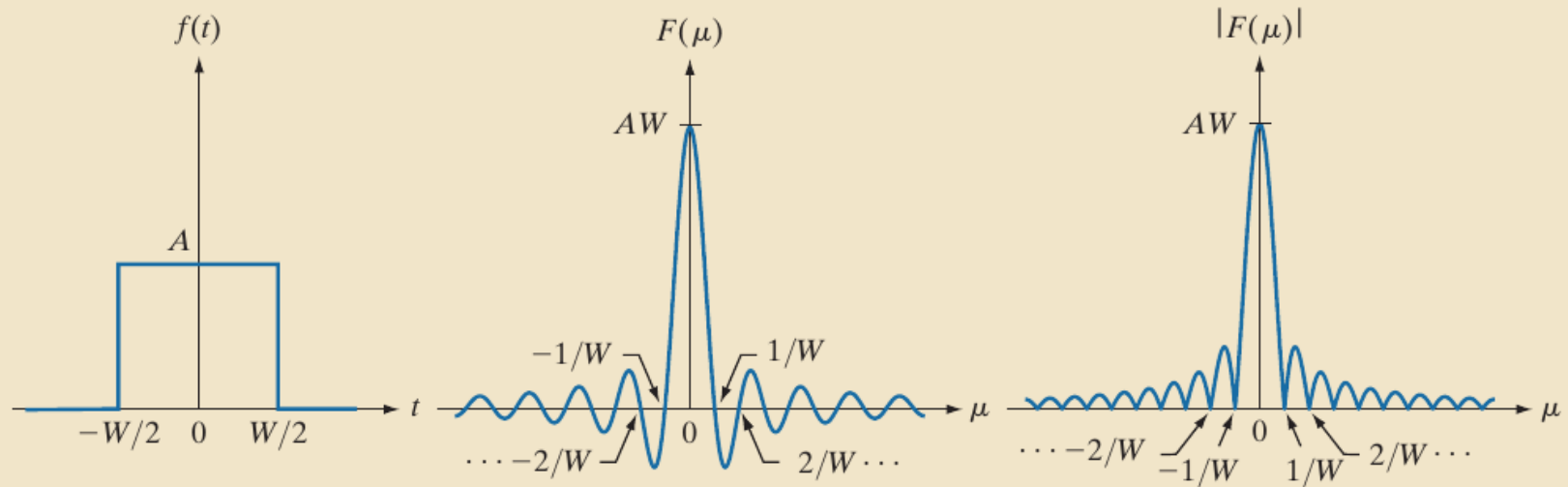


a b
c d

FIGURE 4.3

(a) Continuous impulse located at $t = t_0$. (b) An impulse train consisting of continuous impulses. (c) Unit discrete impulse located at $x = x_0$. (d) An impulse train consisting of discrete unit impulses.





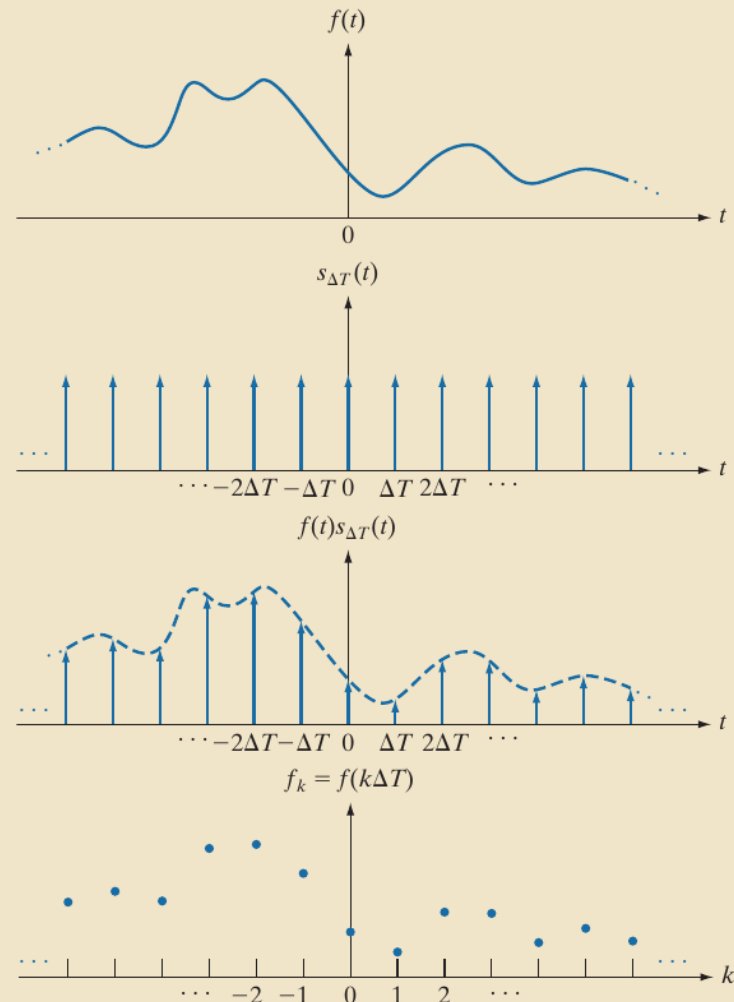
a b c

FIGURE 4.4 (a) A box function, (b) its Fourier transform, and (c) its spectrum. All functions extend to infinity in both directions. Note the inverse relationship between the width, W , of the function and the zeros of the transform.

a
b
c
d

FIGURE 4.5

(a) A continuous function. (b) Train of impulses used to model sampling. (c) Sampled function formed as the product of (a) and (b). (d) Sample values obtained by integration and using the sifting property of impulses. (The dashed line in (c) is shown for reference. It is not part of the data.)



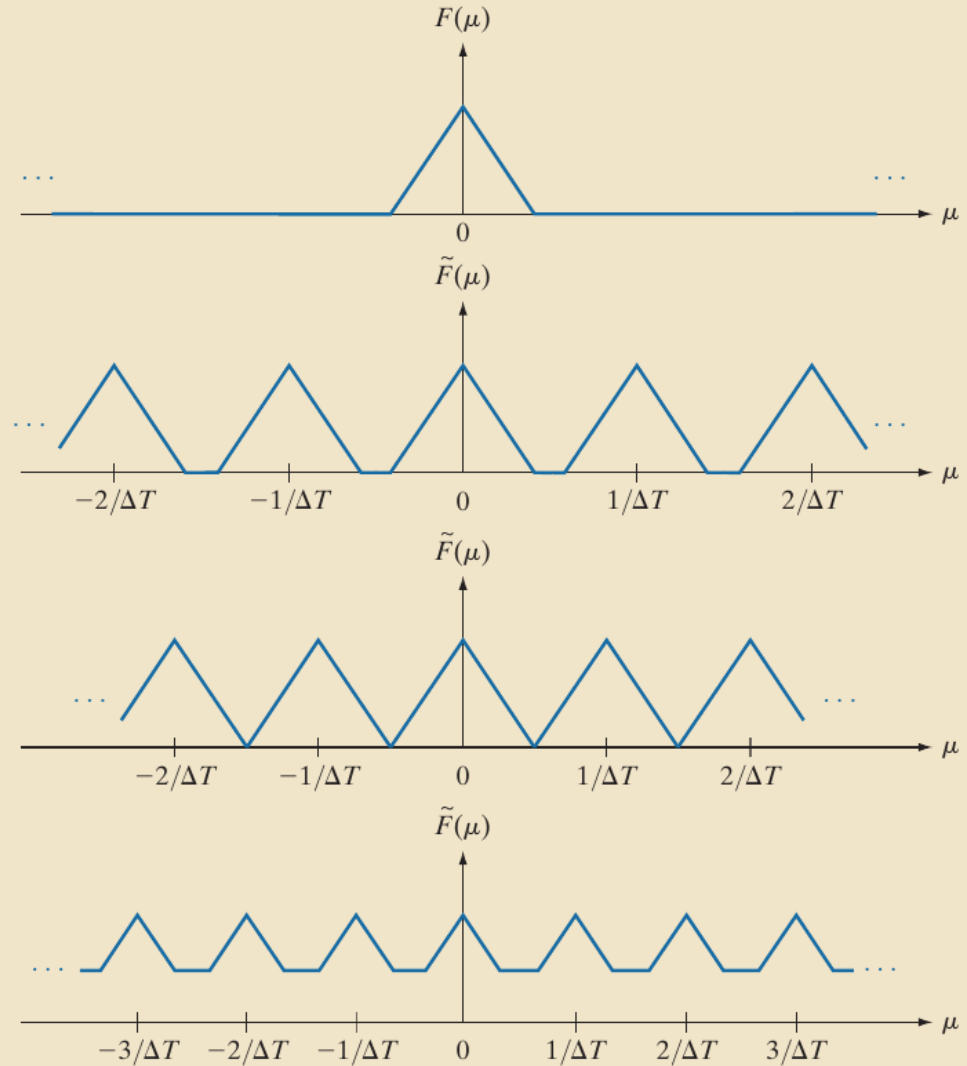
采样(频域)

7/61

a
b
c
d

FIGURE 4.6

(a) Illustrative sketch of the Fourier transform of a band-limited function.
(b)–(d) Transforms of the corresponding sampled functions under the conditions of over-sampling, critically sampling, and under-sampling, respectively.

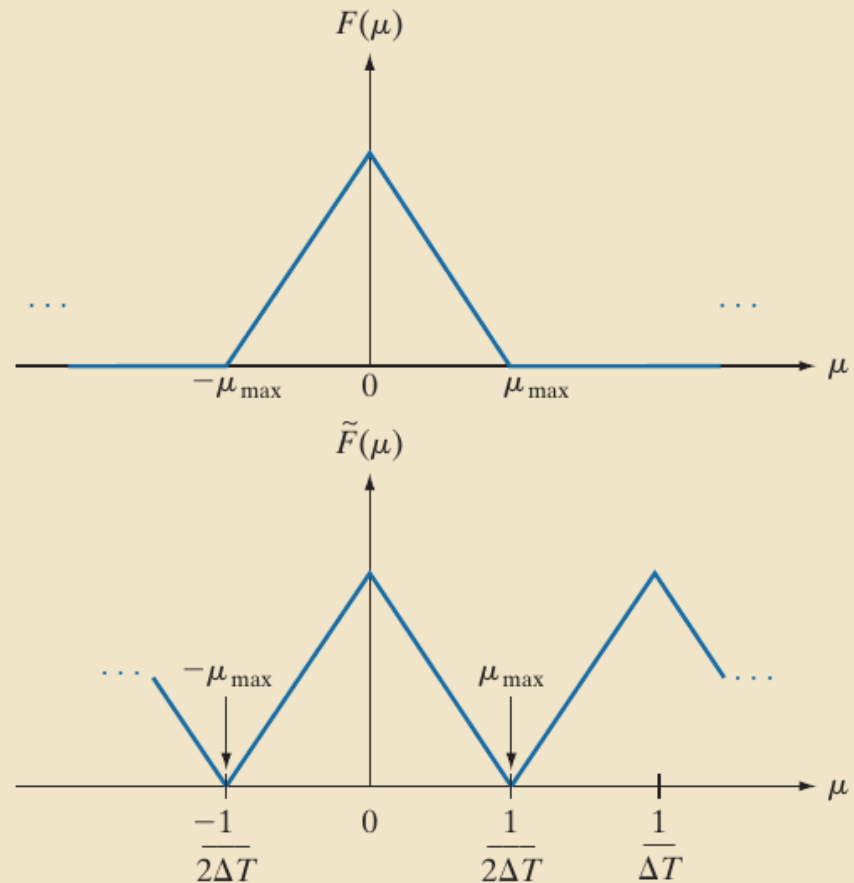


a
b

FIGURE 4.7

(a) Illustrative sketch of the Fourier transform of a band-limited function.

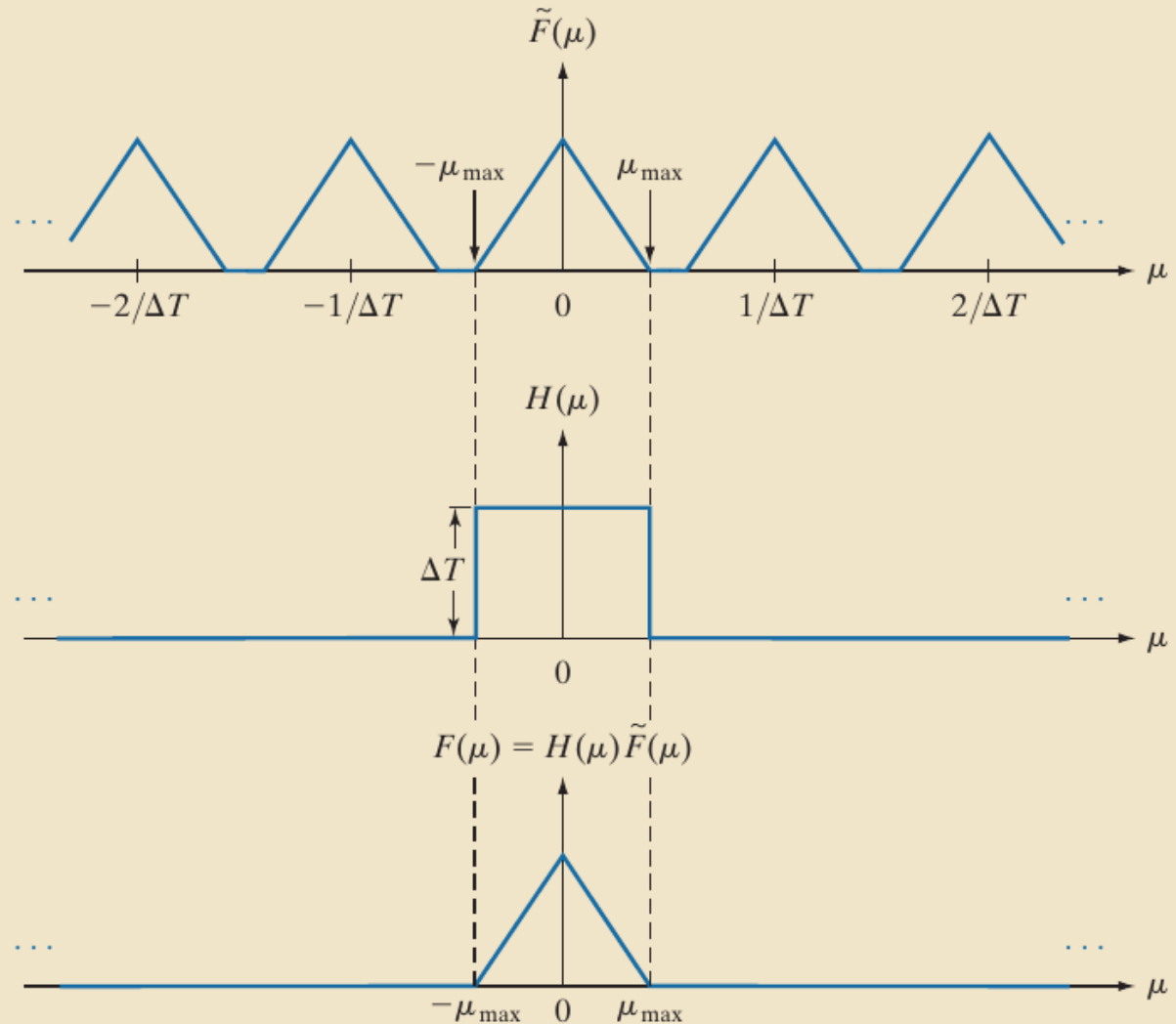
(b) Transform resulting from critically sampling that band-limited function.



a
b
c

FIGURE 4.8

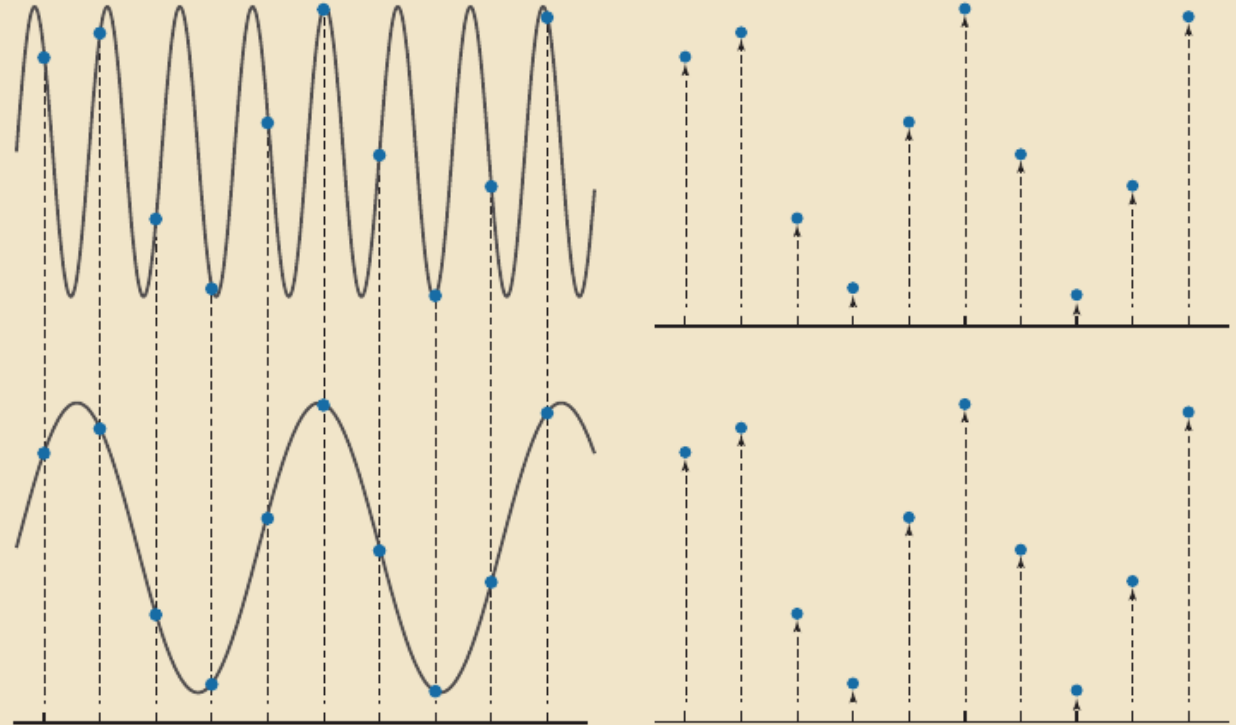
(a) Fourier transform of a sampled, band-limited function.
 (b) Ideal lowpass filter transfer function.
 (c) The product of (b) and (a), used to extract one period of the infinitely periodic sequence in (a).

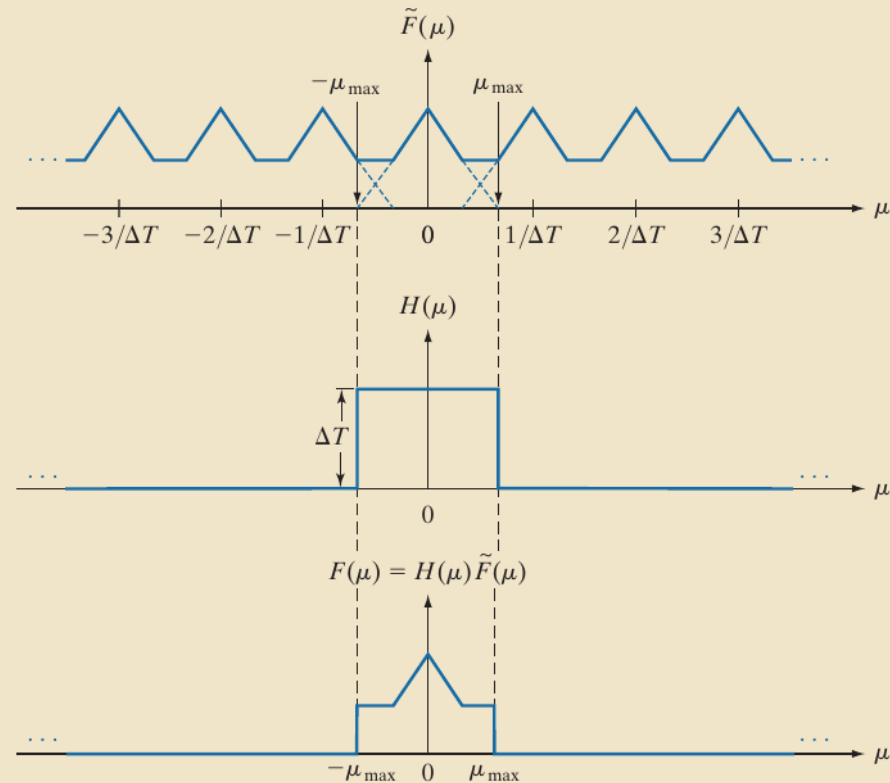


a	b
c	d

FIGURE 4.9

The functions in (a) and (c) are totally different, but their digitized versions in (b) and (d) are identical. Aliasing occurs when the samples of two or more functions coincide, but the functions are different elsewhere.





a
b
c

FIGURE 4.10 (a) Fourier transform of an under-sampled, band-limited function. (Interference between adjacent periods is shown dashed). (b) The same ideal lowpass filter used in Fig. 4.8. (c) The product of (a) and (b). The interference from adjacent periods results in aliasing that prevents perfect recovery of $F(\mu)$ and, consequently, of $f(t)$.

a b

FIGURE 4.12

(a) A continuous function sampled ΔT units apart.

(b) Samples in the x -domain.

Variable t is continuous, while x is discrete.

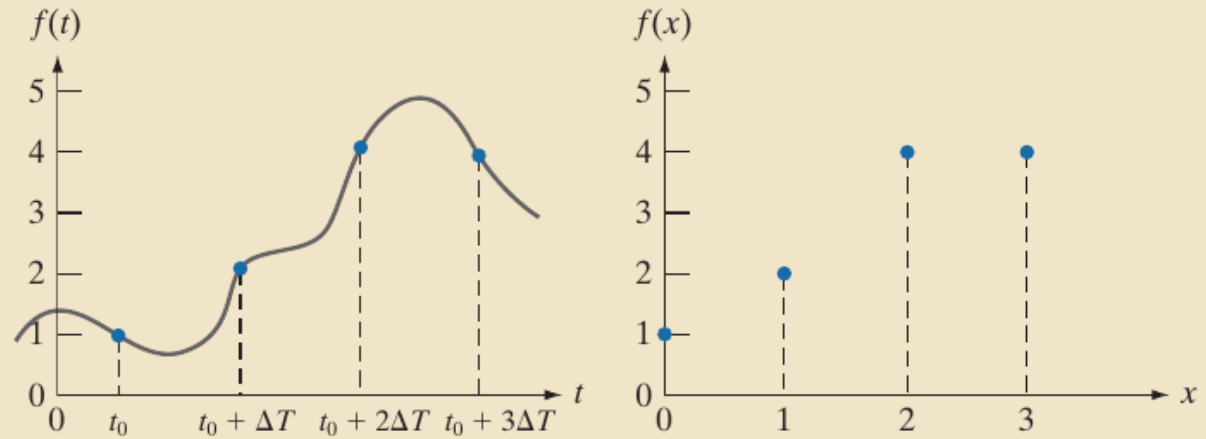
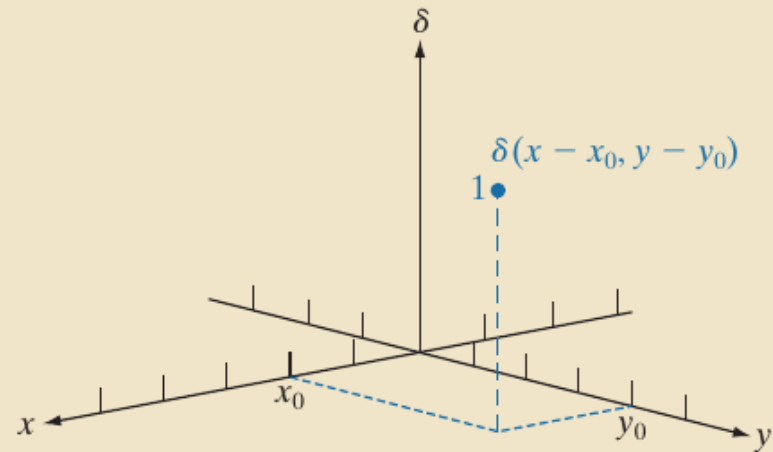


FIGURE 4.13

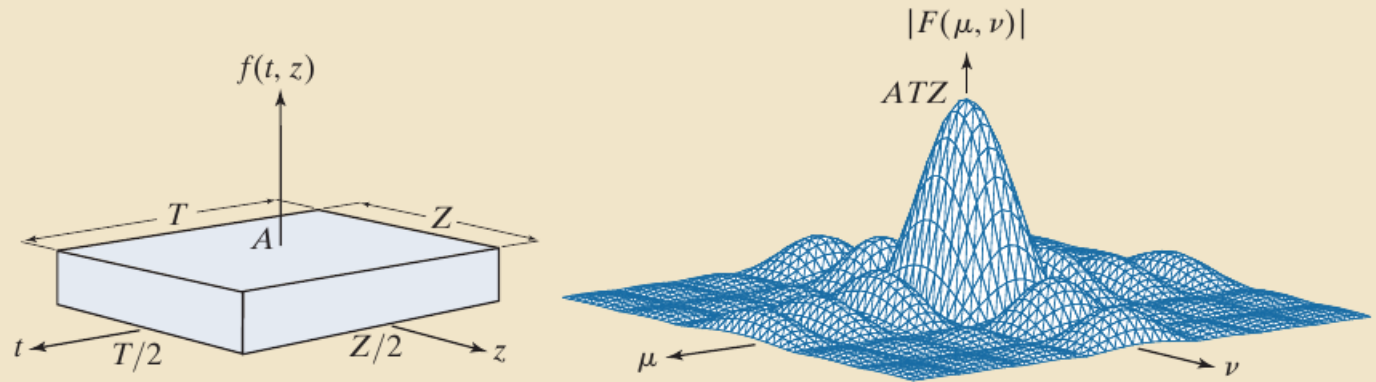
2-D unit discrete impulse. Variables x and y are discrete, and δ is zero everywhere except at coordinates (x_0, y_0) , where its value is 1.



a b

FIGURE 4.14

(a) A 2-D function and (b) a section of its spectrum. The box is longer along the t -axis, so the spectrum is more contracted along the μ -axis.



a b

FIGURE 4.16

Two-dimensional Fourier transforms of (a) an over-sampled, and (b) an under-sampled, band-limited function.

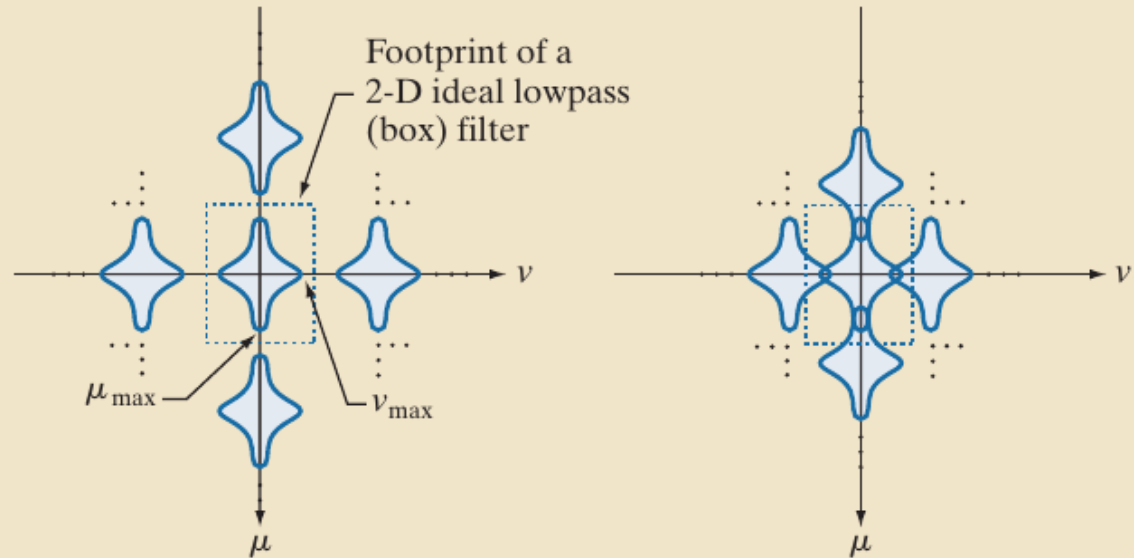
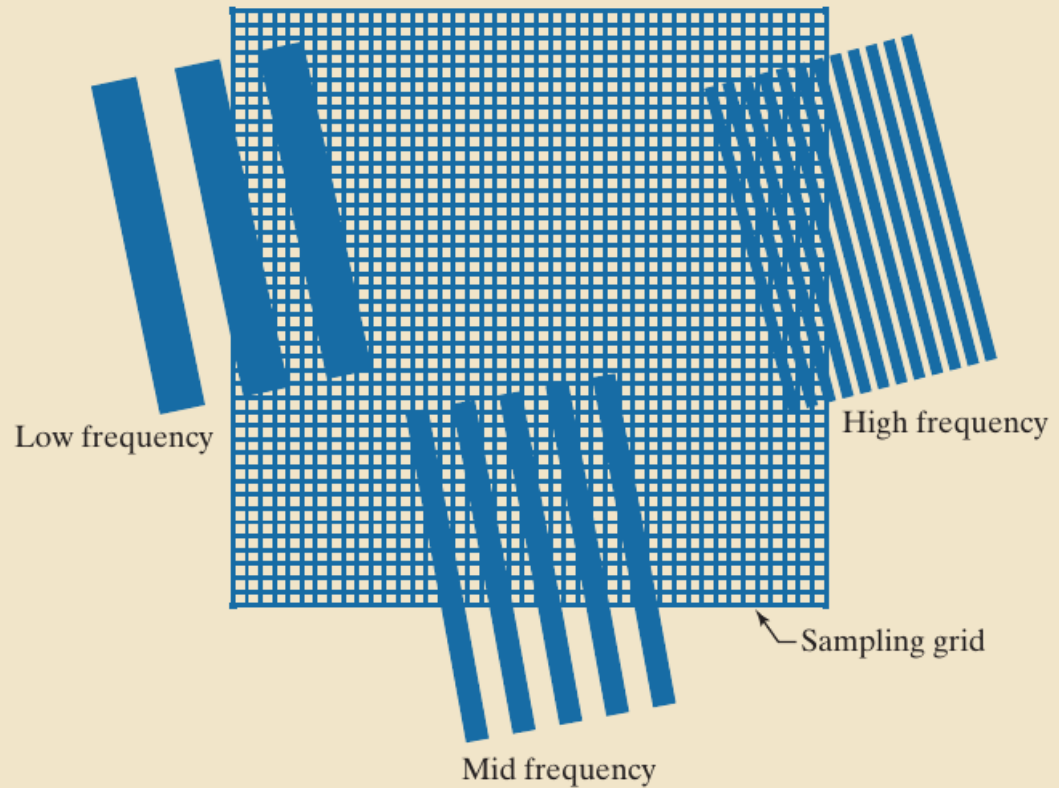


FIGURE 4.17

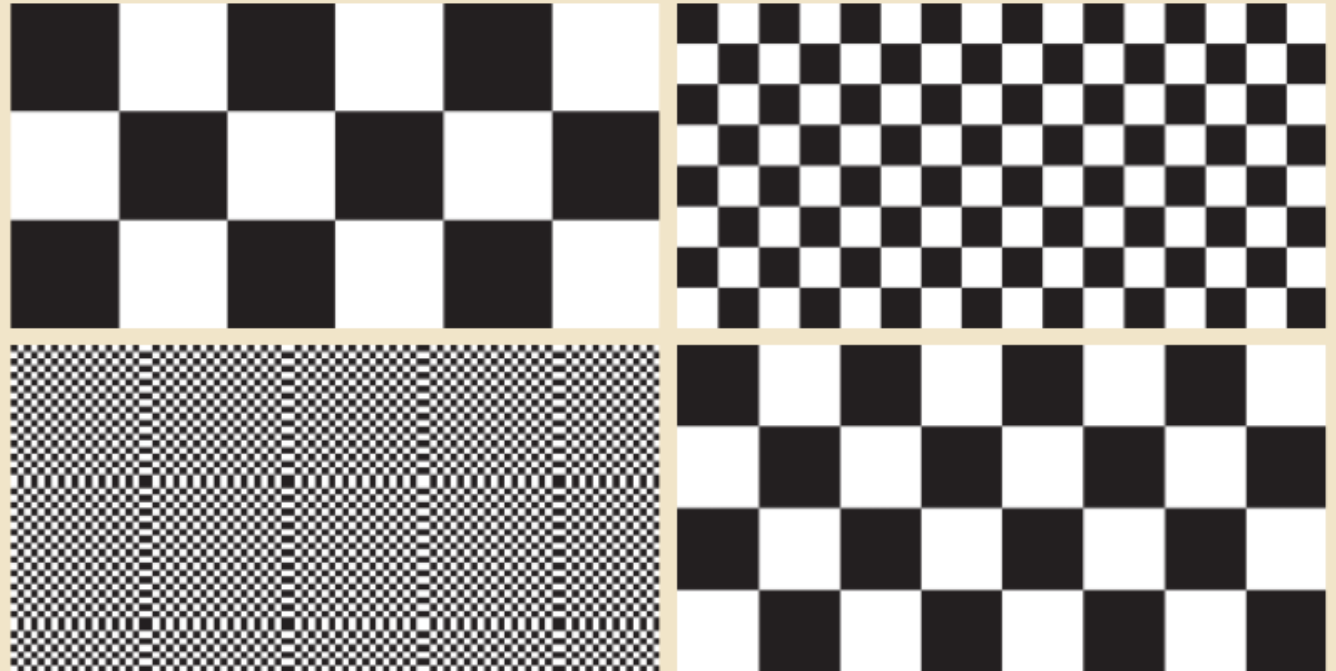
Various aliasing effects resulting from the interaction between the frequency of 2-D signals and the sampling rate used to digitize them. The regions outside the sampling grid are continuous and free of aliasing.



a b
c d

FIGURE 4.18

Aliasing. In (a) and (b) the squares are of sizes 16 and 6 pixels on the side. In (c) and (d) the squares are of sizes 0.95 and 0.48 pixels, respectively. Each small square in (c) is one pixel. Both (c) and (d) are aliased. Note how (d) masquerades as a “normal” image.





a b c

FIGURE 4.19 Illustration of aliasing on resampled natural images. (a) A digital image of size 772×548 pixels with visually negligible aliasing. (b) Result of resizing the image to 33% of its original size by pixel deletion and then restoring it to its original size by pixel replication. Aliasing is clearly visible. (c) Result of blurring the image in (a) with an averaging filter prior to resizing. The image is slightly more blurred than (b), but aliasing is not longer objectionable. (Original image courtesy of the Signal Compression Laboratory, University of California, Santa Barbara.)

a	b	c
d	e	f

FIGURE 4.20

Examples of the moiré effect. These are vector drawings, not digitized patterns. Superimposing one pattern on the other is analogous to multiplying the patterns.

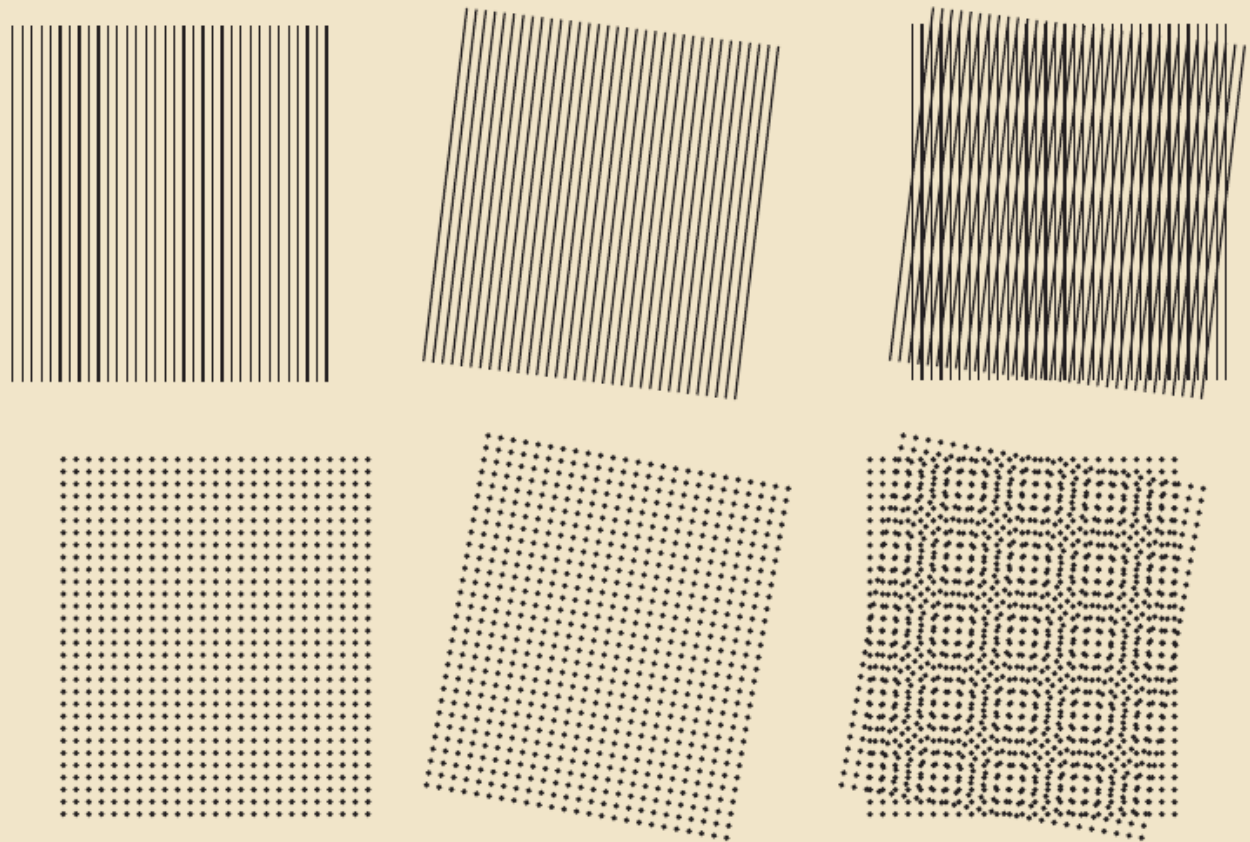
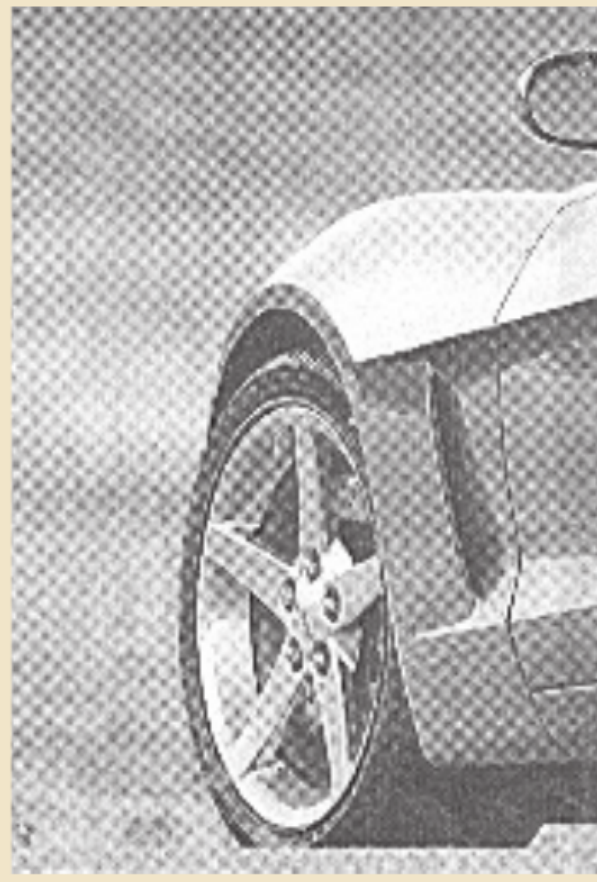


FIGURE 4.21

A newspaper image digitized at 75 dpi. Note the moiré-like pattern resulting from the interaction between the $\pm 45^\circ$ orientation of the half-tone dots and the north-south orientation of the sampling elements used to digitized the image.

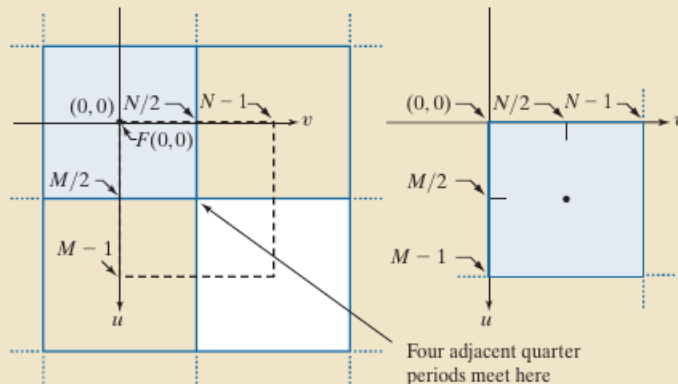
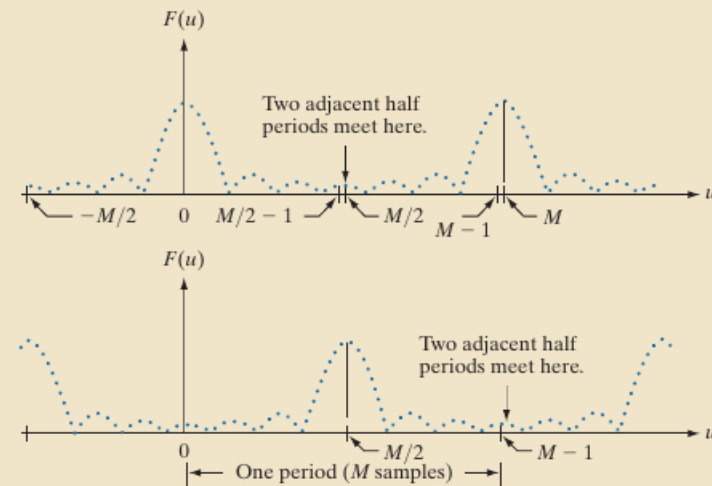


a
b
c d

FIGURE 4.22

Centering the Fourier transform.

(a) A 1-D DFT showing an infinite number of periods. (b) Shifted DFT obtained by multiplying $f(x)$ by $(-1)^x$ before computing $F(u)$. (c) A 2-D DFT showing an infinite number of periods. The area within the dashed rectangle is the data array, $F(u, v)$, obtained with Eq. (4-67) with an image $f(x, y)$ as the input. This array consists of four quarter periods. (d) Shifted array obtained by multiplying $f(x, y)$ by $(-1)^{x+y}$ before computing $F(u, v)$. The data now contains one complete, centered period, as in (b).



= $M \times N$ data array computed by the DFT with $f(x, y)$ as input

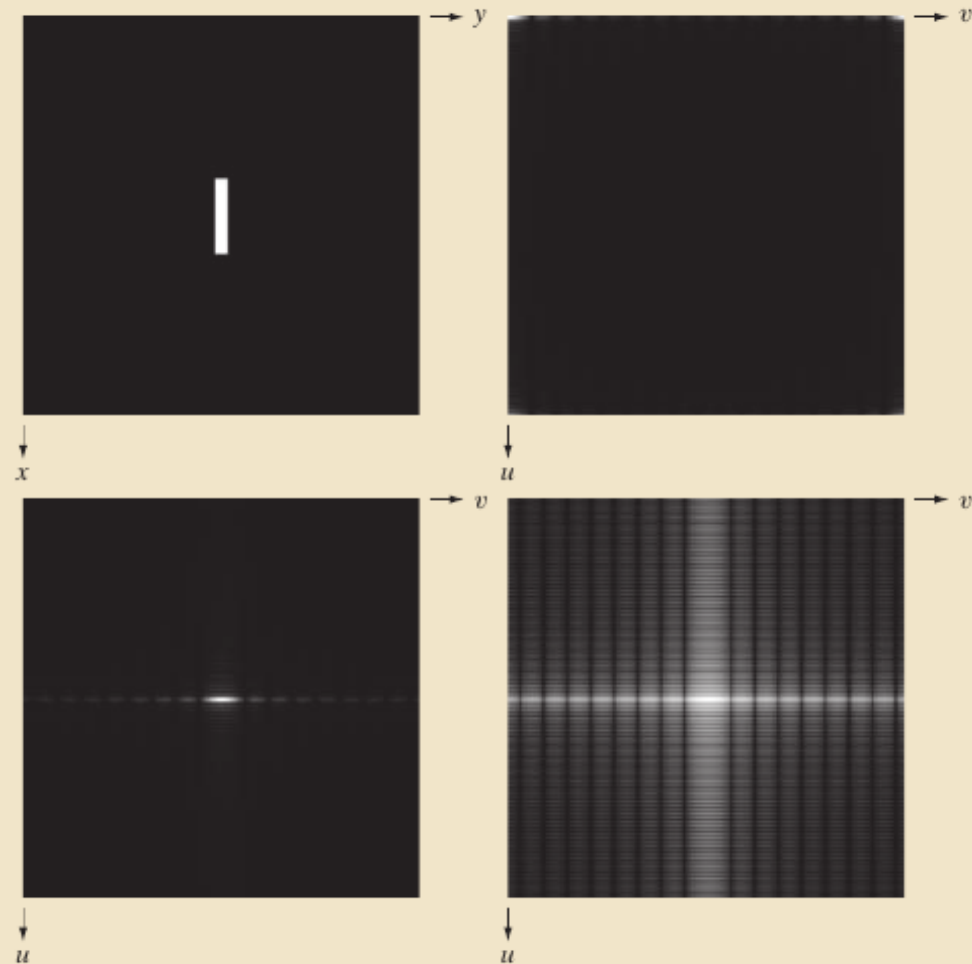
= $M \times N$ data array computed by the DFT with $f(x, y)(-1)^{x+y}$ as input

= Periods of the DFT

a b
c d

FIGURE 4.23

(a) Image.
(b) Spectrum, showing small, bright areas in the four corners (you have to look carefully to see them).
(c) Centered spectrum.
(d) Result after a log transformation. The zero crossings of the spectrum are closer in the vertical direction because the rectangle in (a) is longer in that direction. The right-handed coordinate convention used in the book places the origin of the spatial and frequency domains at the top left (see Fig. 2.19).



a b
c d

FIGURE 4.24

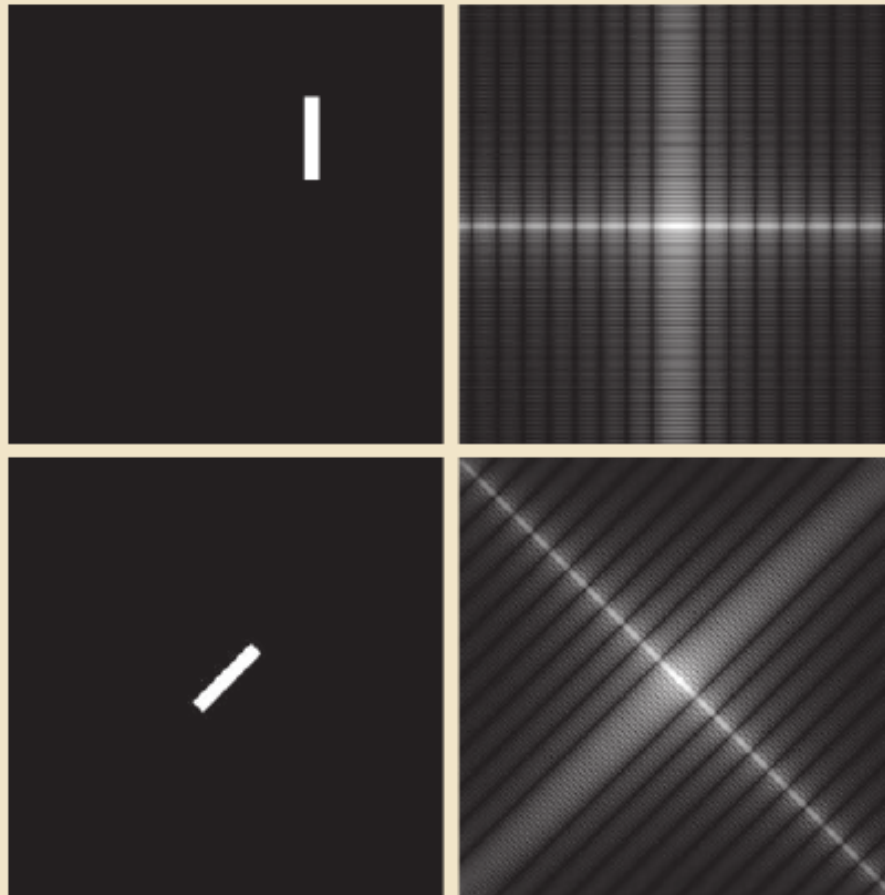
(a) The rectangle in Fig. 4.23(a) translated.

(b) Corresponding spectrum.

(c) Rotated rectangle.

(d) Corresponding spectrum.

The spectrum of the translated rectangle is identical to the spectrum of the original image in Fig. 4.23(a).



a b c

FIGURE 4.25
Phase angle
images of
(a) centered,
(b) translated,
and (c) rotated
rectangles.

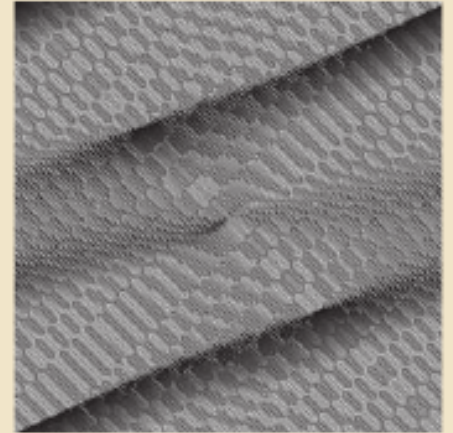
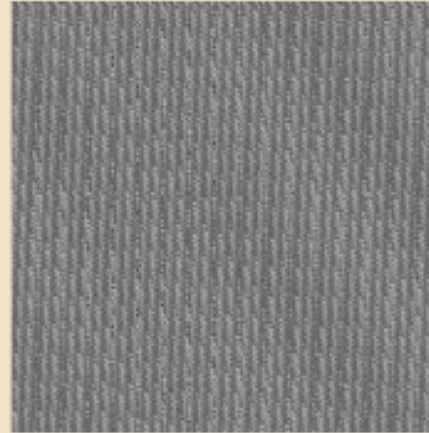


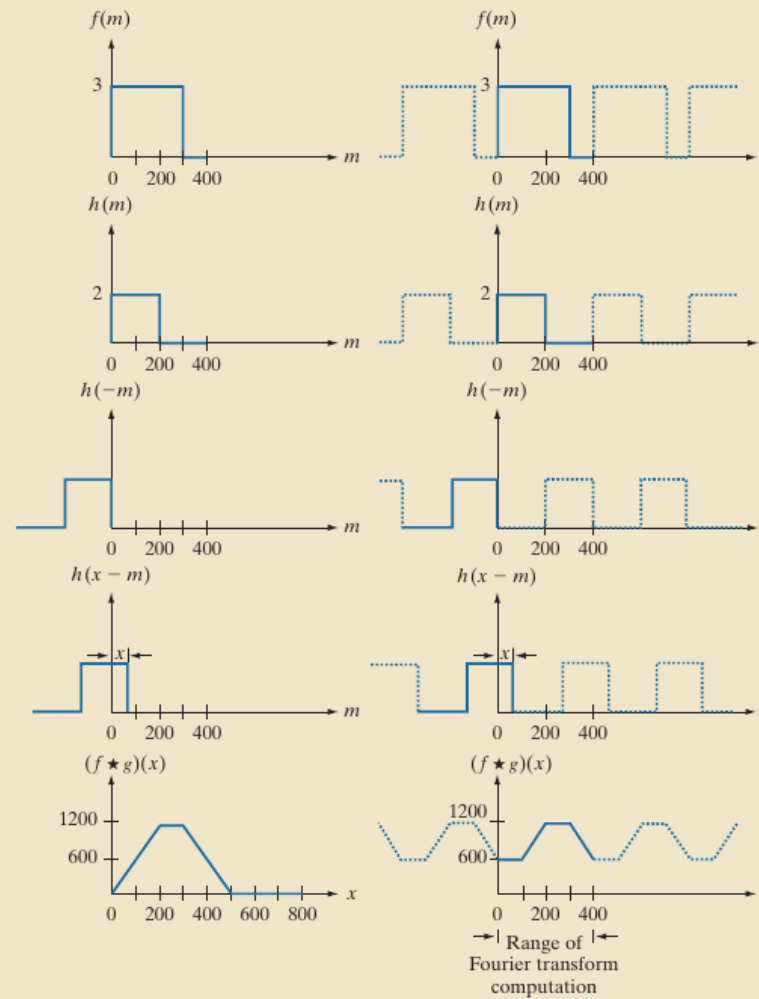


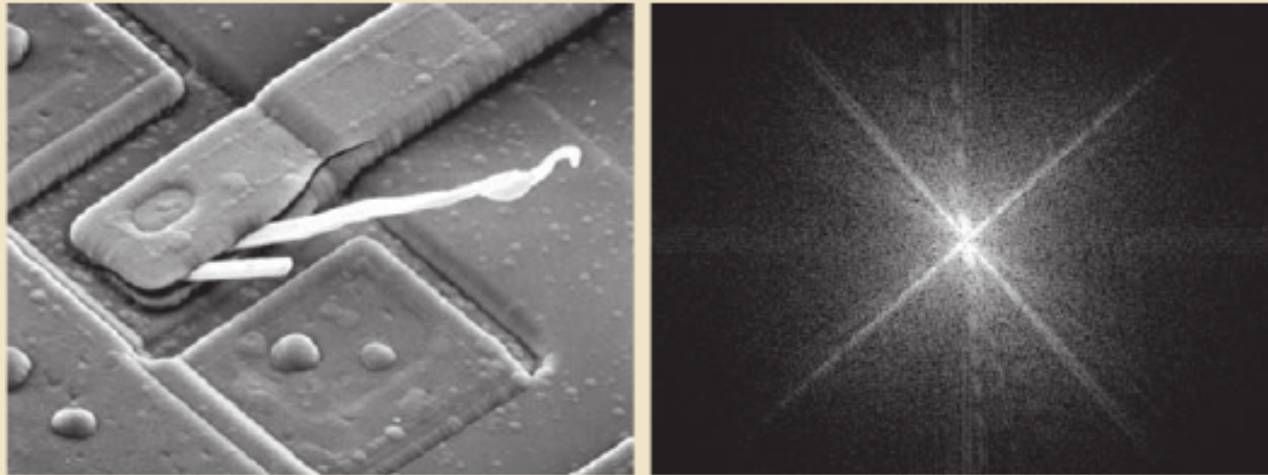
FIGURE 4.26 (a) Boy image. (b) Phase angle. (c) Boy image reconstructed using only its phase angle (all shape features are there, but the intensity information is missing because the spectrum was not used in the reconstruction). (d) Boy image reconstructed using only its spectrum. (e) Boy image reconstructed using its phase angle and the spectrum of the rectangle in Fig. 4.23(a). (f) Rectangle image reconstructed using its phase and the spectrum of the boy's image.

a f
b g
c h
d i
e j

FIGURE 4.27

Left column:
Spatial
convolution
computed with
Eq. (3-35), using
the approach
discussed in
Section 3.4.
Right column:
Circular
convolution. The
solid line in (j)
is the result we
would obtain
using the DFT,
or, equivalently,
Eq. (4-48). This
erroneous result
can be remedied
by using zero
padding.



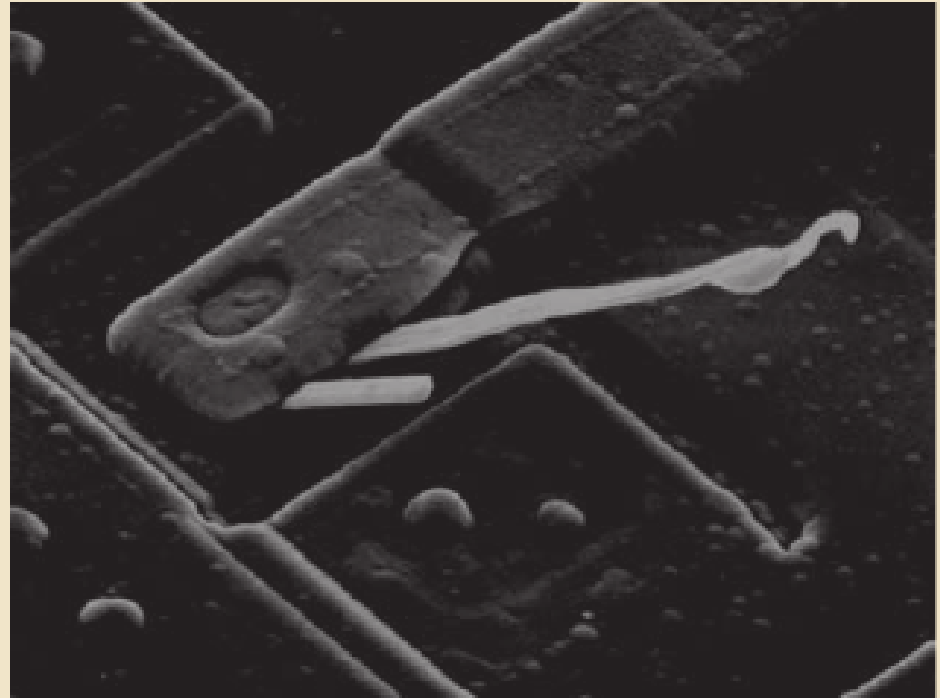


a b

FIGURE 4.28 (a) SEM image of a damaged integrated circuit. (b) Fourier spectrum of (a). (Original image courtesy of Dr. J. M. Hudak, Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, Canada.)

FIGURE 4.29

Result of filtering the image in Fig. 4.28(a) with a filter transfer function that sets to 0 the dc term, $F(P/2, Q/2)$, in the centered Fourier transform, while leaving all other transform terms unchanged.



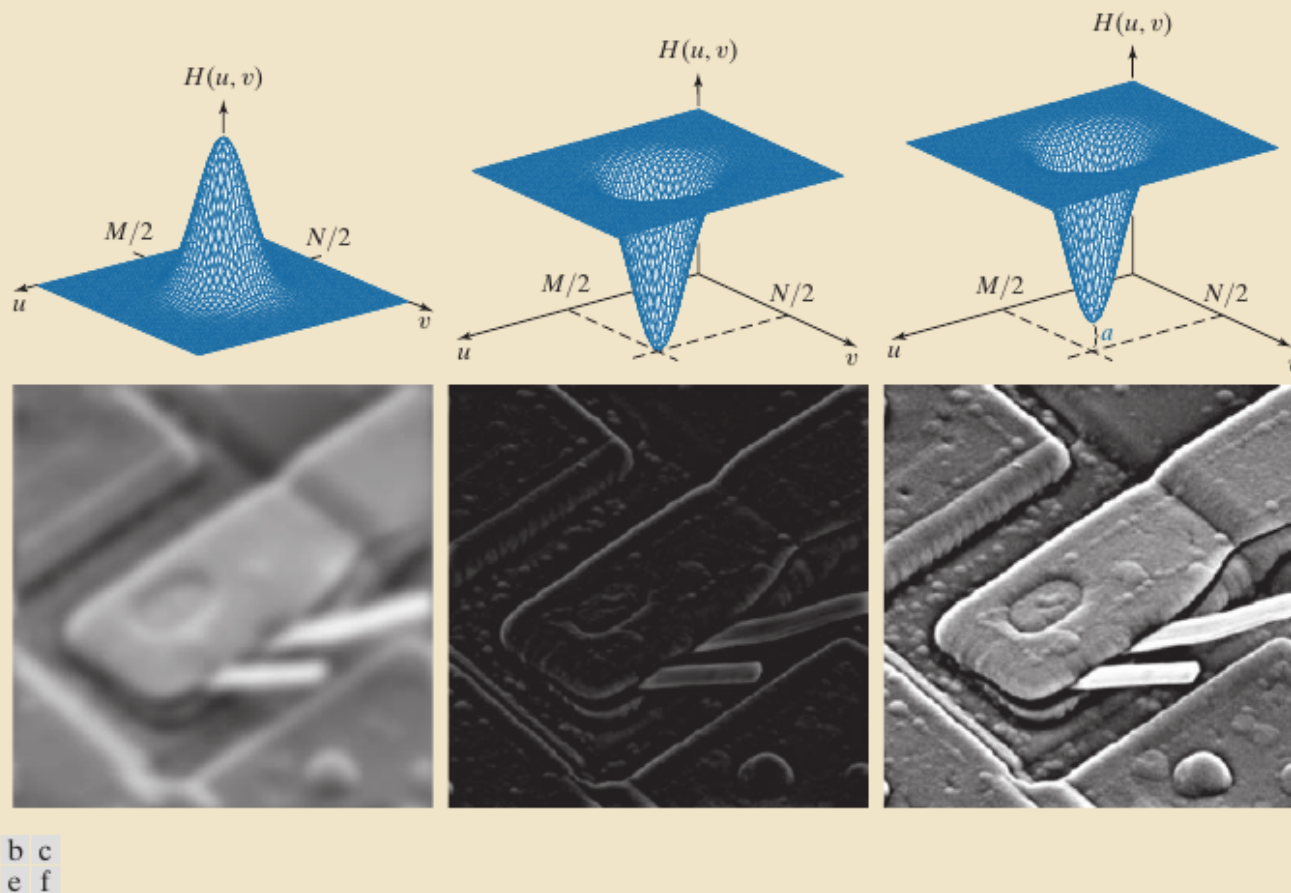
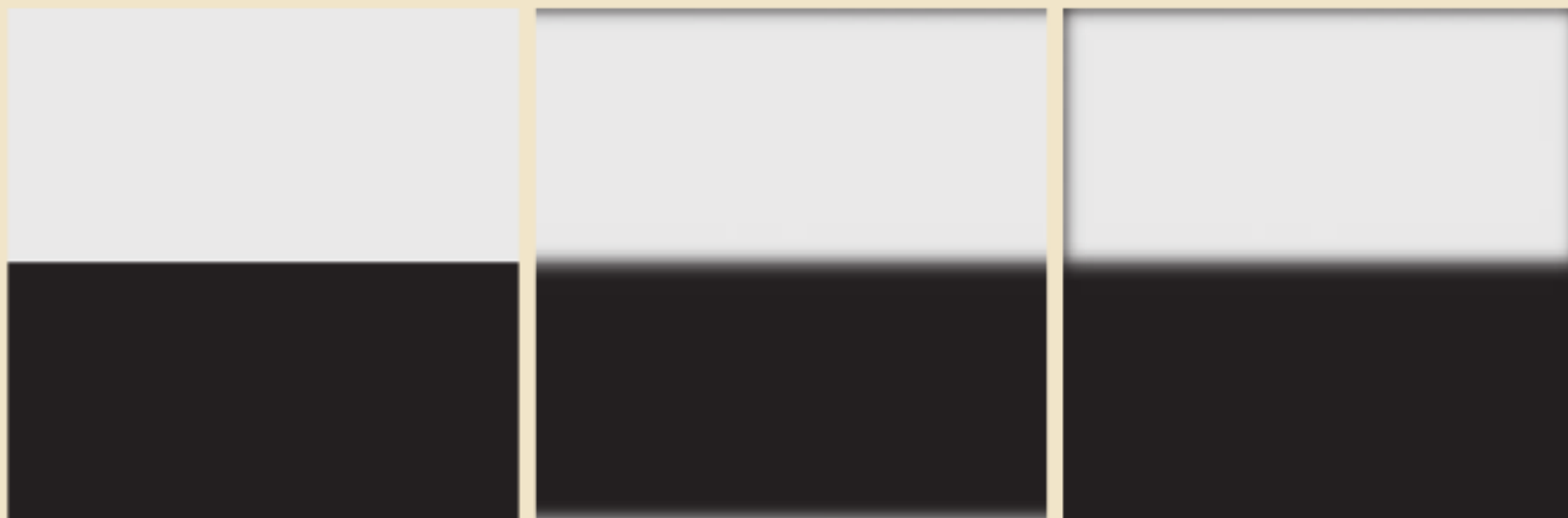
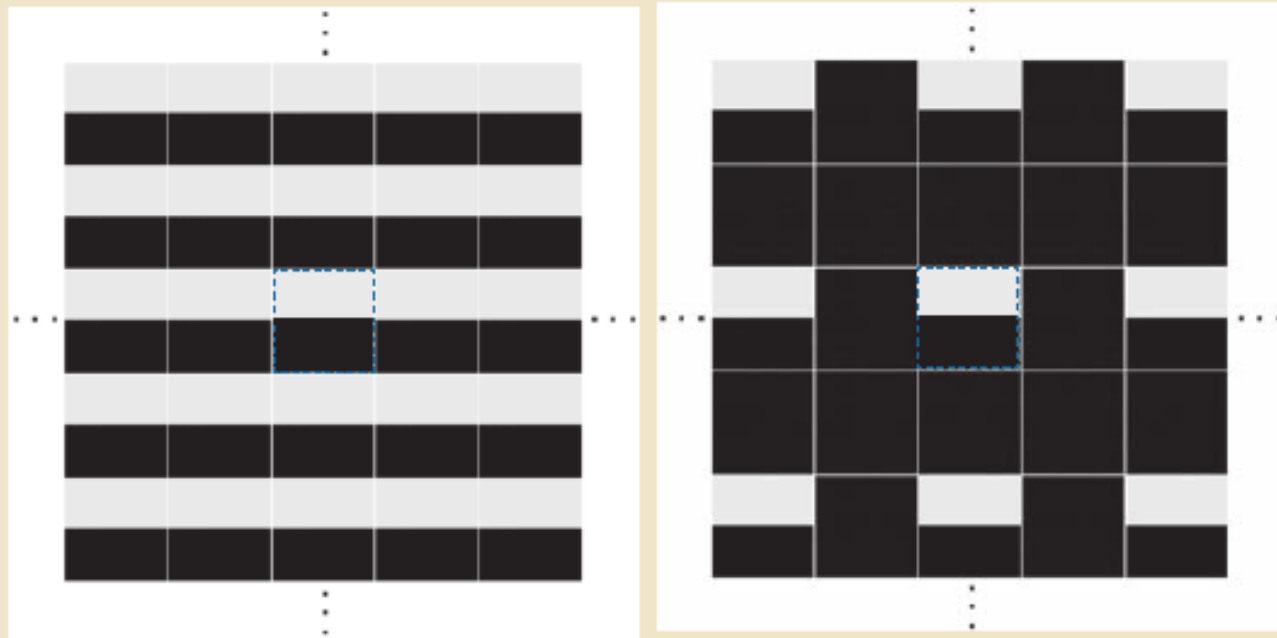


FIGURE 4.30 Top row: Frequency domain filter transfer functions of (a) a lowpass filter, (b) a highpass filter, and (c) an offset highpass filter. Bottom row: Corresponding filtered images obtained using Eq. (4-104). The offset in (c) is $a = 0.85$, and the height of $H(u, v)$ is 1. Compare (f) with Fig. 4.28(a).



a b c

FIGURE 4.31 (a) A simple image. (b) Result of blurring with a Gaussian lowpass filter without padding. (c) Result of lowpass filtering with zero padding. Compare the vertical edges in (b) and (c).



a b

FIGURE 4.32 (a) Image periodicity without image padding. (b) Periodicity after padding with 0's (black). The dashed areas in the center correspond to the image in Fig. 4.31(a). Periodicity is inherent when using the DFT. (The thin white lines in both images are superimposed for clarity; they are not part of the data.)

a c
b d

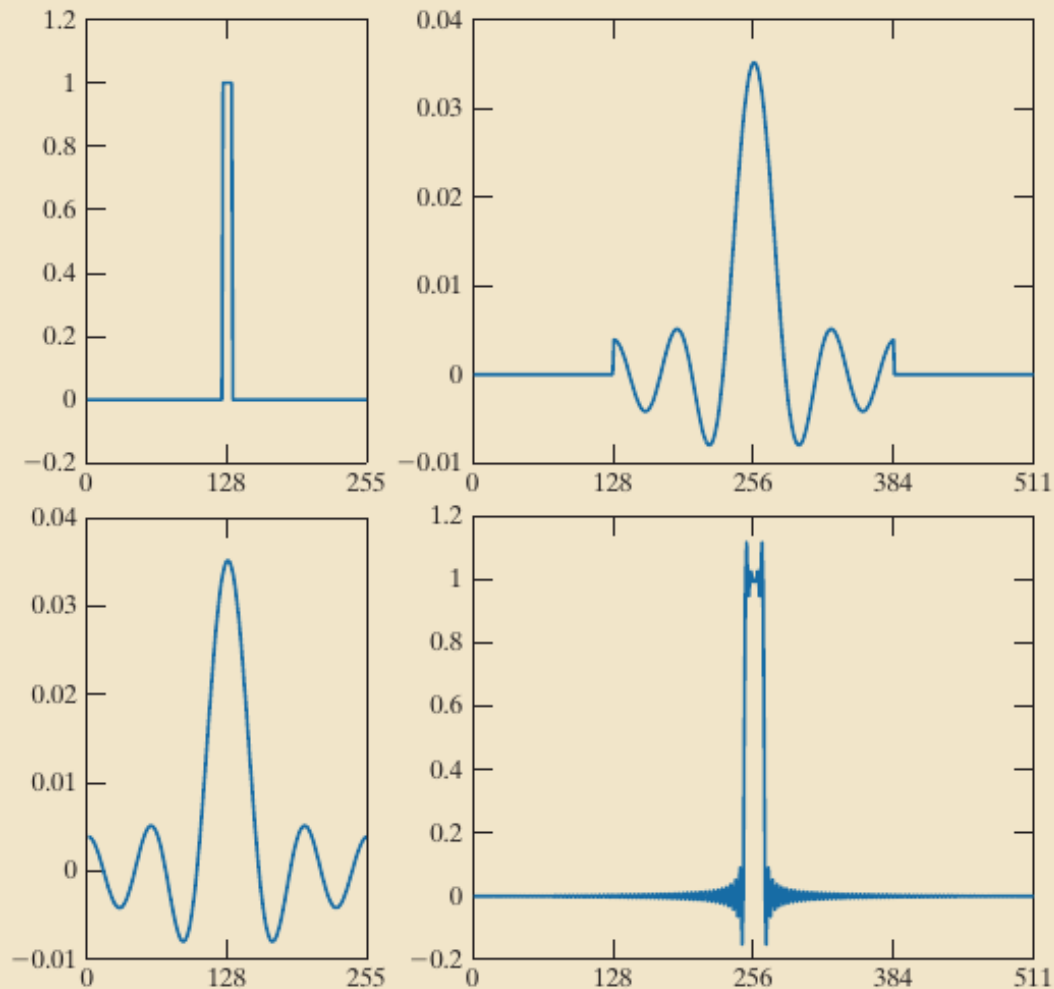
FIGURE 4.33

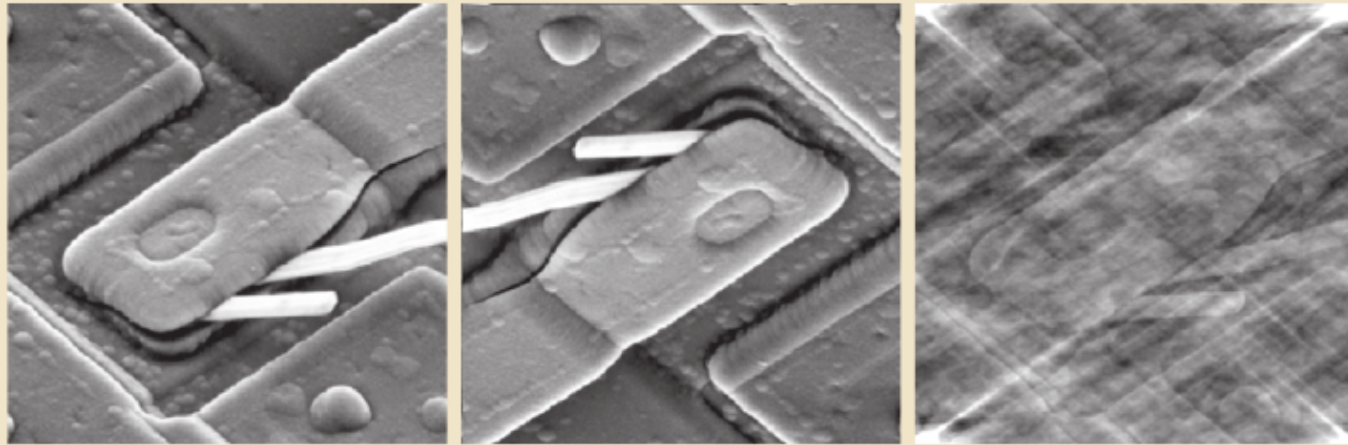
(a) Filter transfer function specified in the (centered) frequency domain.

(b) Spatial representation (filter kernel) obtained by computing the IDFT of (a).

(c) Result of padding (b) to twice its length (note the discontinuities).

(d) Corresponding filter in the frequency domain obtained by computing the DFT of (c). Note the ringing caused by the discontinuities in (c). Part (b) of the figure is below (a), and (d) is below (c).





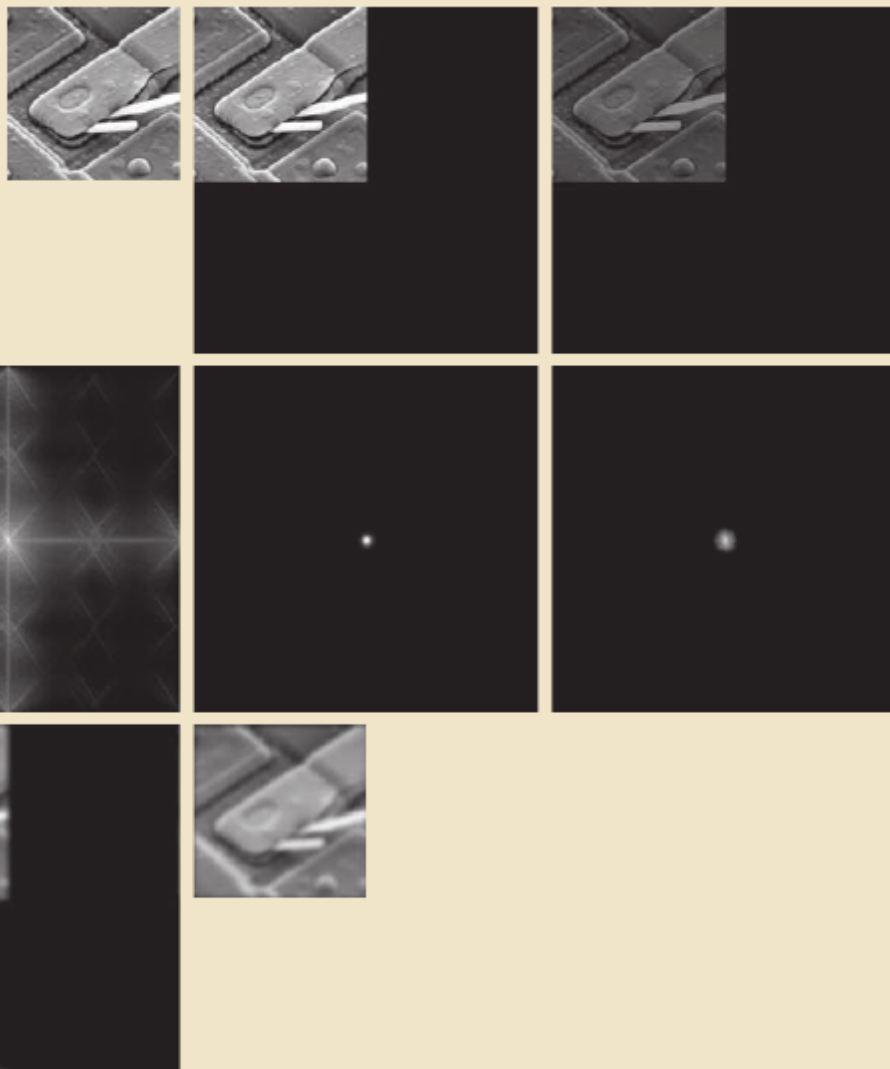
a b c

FIGURE 4.34 (a) Original image. (b) Image obtained by multiplying the phase angle array by -1 in Eq. (4-86) and computing the IDFT. (c) Result of multiplying the phase angle by 0.25 and computing the IDFT. The magnitude of the transform, $|F(u, v)|$, used in (b) and (c) was the same.

a b c
d e f
g h

FIGURE 4.35

- (a) An $M \times N$ image, f .
(b) Padded image, f_p , of size $P \times Q$.
(c) Result of multiplying f_p by $(-1)^{x+y}$.
(d) Spectrum of F . (e) Centered Gaussian lowpass filter transfer function, H , of size $P \times Q$.
(f) Spectrum of the product HF .
(g) Image g_p , the real part of the IDFT of HF , multiplied by $(-1)^{x+y}$.
(h) Final result, g , obtained by extracting the first M rows and N columns of g_p .



a b
c d

-1	0	1
-2	0	2
-1	0	1

FIGURE 4.38

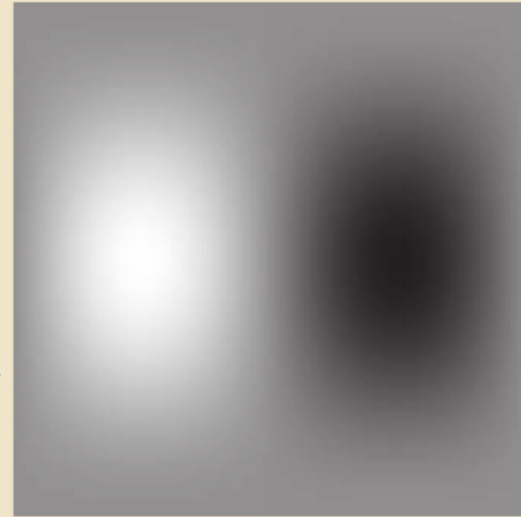
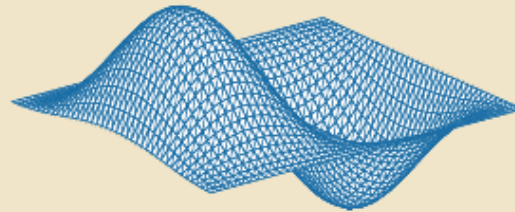
(a) A spatial kernel and perspective plot of its corresponding frequency domain filter transfer function.

(b) Transfer function shown as an image.

(c) Result of filtering

Fig. 4.37(a) in the frequency domain with the transfer function in (b).

(d) Result of filtering the same image in the spatial domain with the kernel in (a). The results are identical.



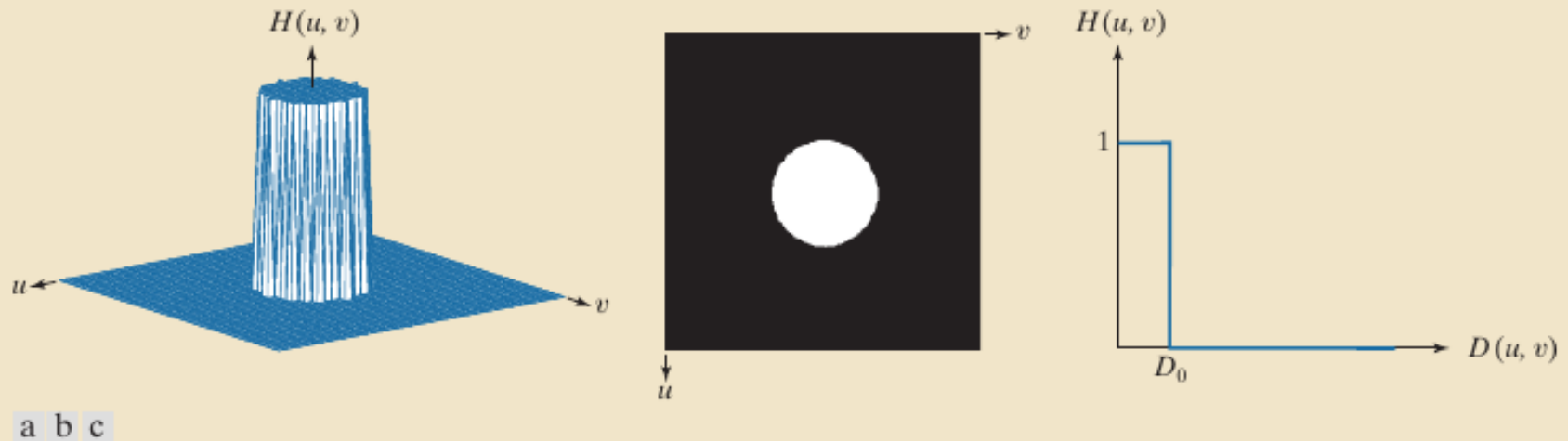
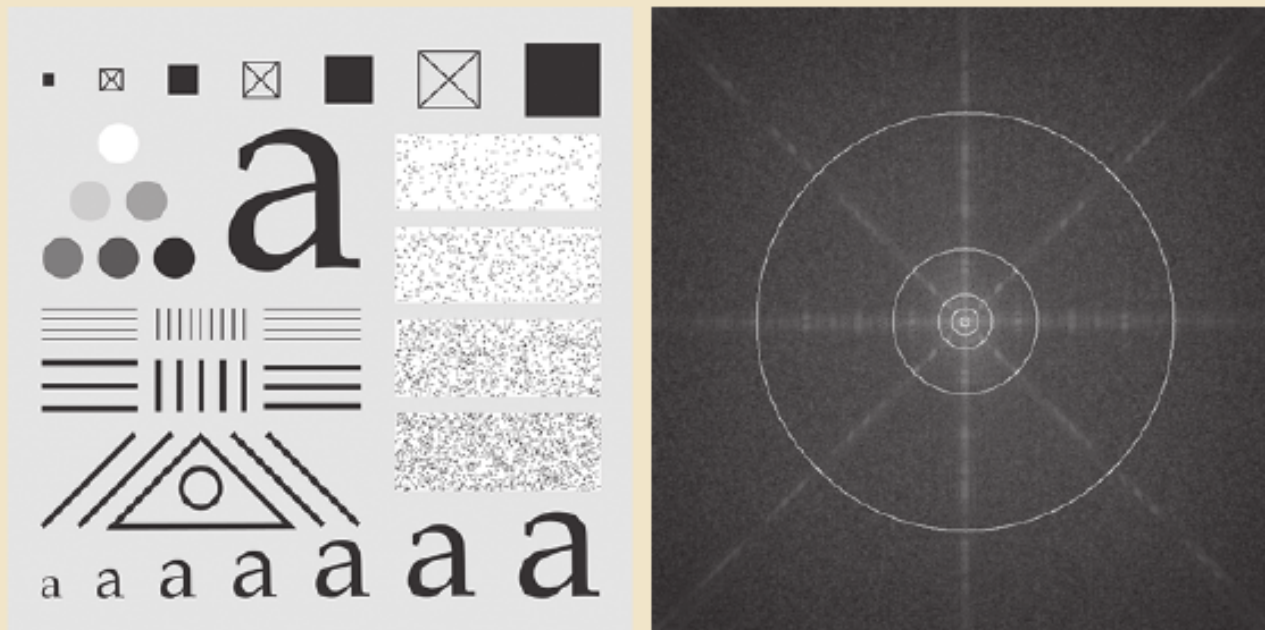
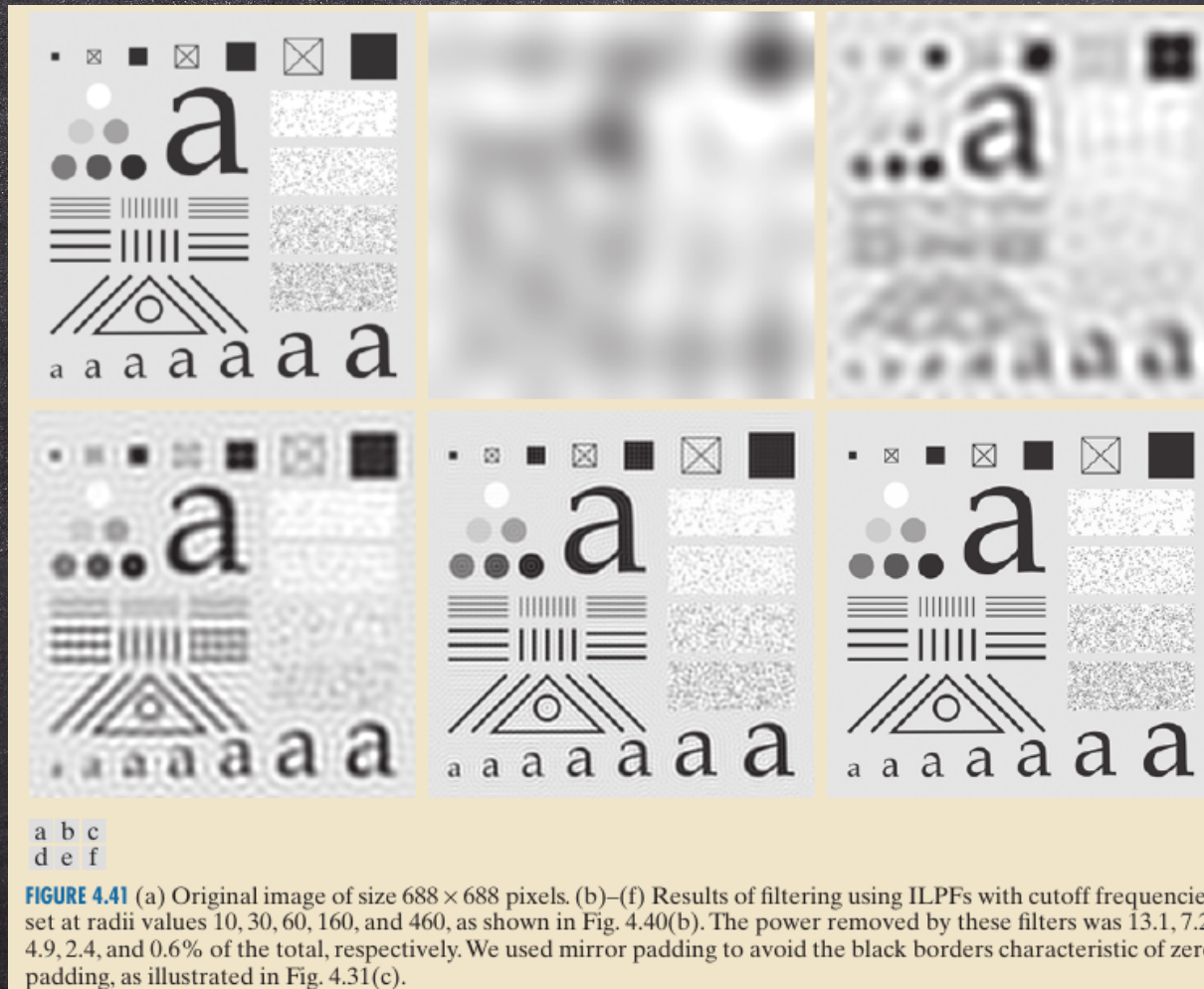


FIGURE 4.39 (a) Perspective plot of an ideal lowpass-filter transfer function. (b) Function displayed as an image. (c) Radial cross section.



a b

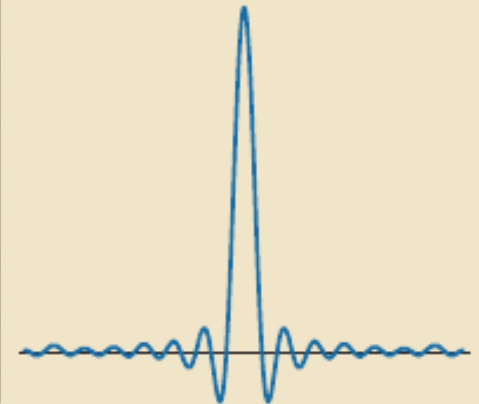
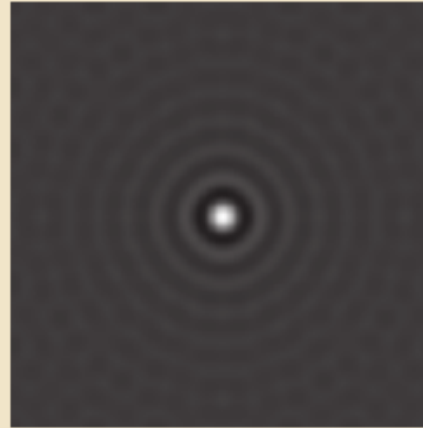
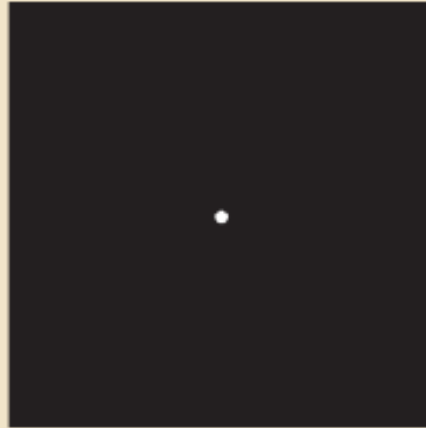
FIGURE 4.40 (a) Test pattern of size 688×688 pixels, and (b) its spectrum. The spectrum is double the image size as a result of padding, but is shown half size to fit. The circles have radii of 10, 30, 60, 160, and 460 pixels with respect to the full-size spectrum. The radii enclose 86.9, 92.8, 95.1, 97.6, and 99.4% of the padded image power, respectively.

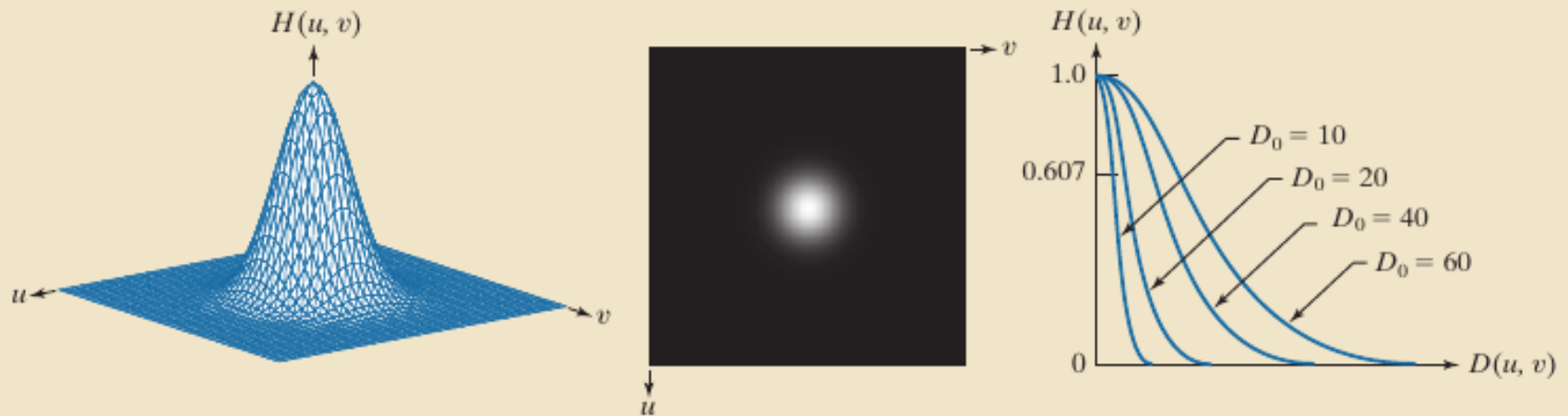


a b c

FIGURE 4.42

(a) Frequency domain ILPF transfer function.
(b) Corresponding spatial domain kernel function.
(c) Intensity profile of a horizontal line through the center of (b).





a b c

FIGURE 4.43 (a) Perspective plot of a GLPF transfer function. (b) Function displayed as an image. (c) Radial cross sections for various values of D_0 .



FIGURE 4.44 (a) Original image of size 688×688 pixels. (b)–(f) Results of filtering using GLPFs with cutoff frequencies at the radii shown in Fig. 4.40. Compare with Fig. 4.41. We used mirror padding to avoid the black borders characteristic of zero padding.

$$H(u, v) = \frac{1}{1 + [D(u, v) / D_0]^{2n}}$$

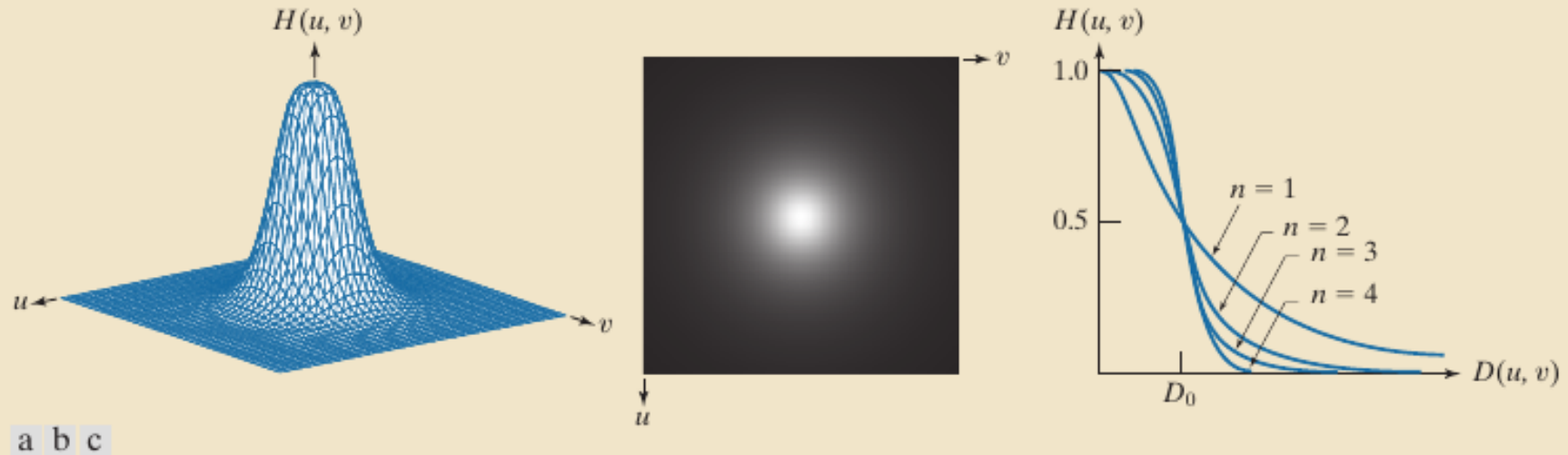


FIGURE 4.45 (a) Perspective plot of a Butterworth lowpass-filter transfer function. (b) Function displayed as an image. (c) Radial cross sections of BLPFs of orders 1 through 4.



FIGURE 4.46 (a) Original image of size 688×688 pixels. (b)–(f) Results of filtering using BLPPFs with cutoff frequencies at the radii shown in Fig. 4.40 and $n = 2.25$. Compare with Figs. 4.41 and 4.44. We used mirror padding to avoid the black borders characteristic of zero padding.

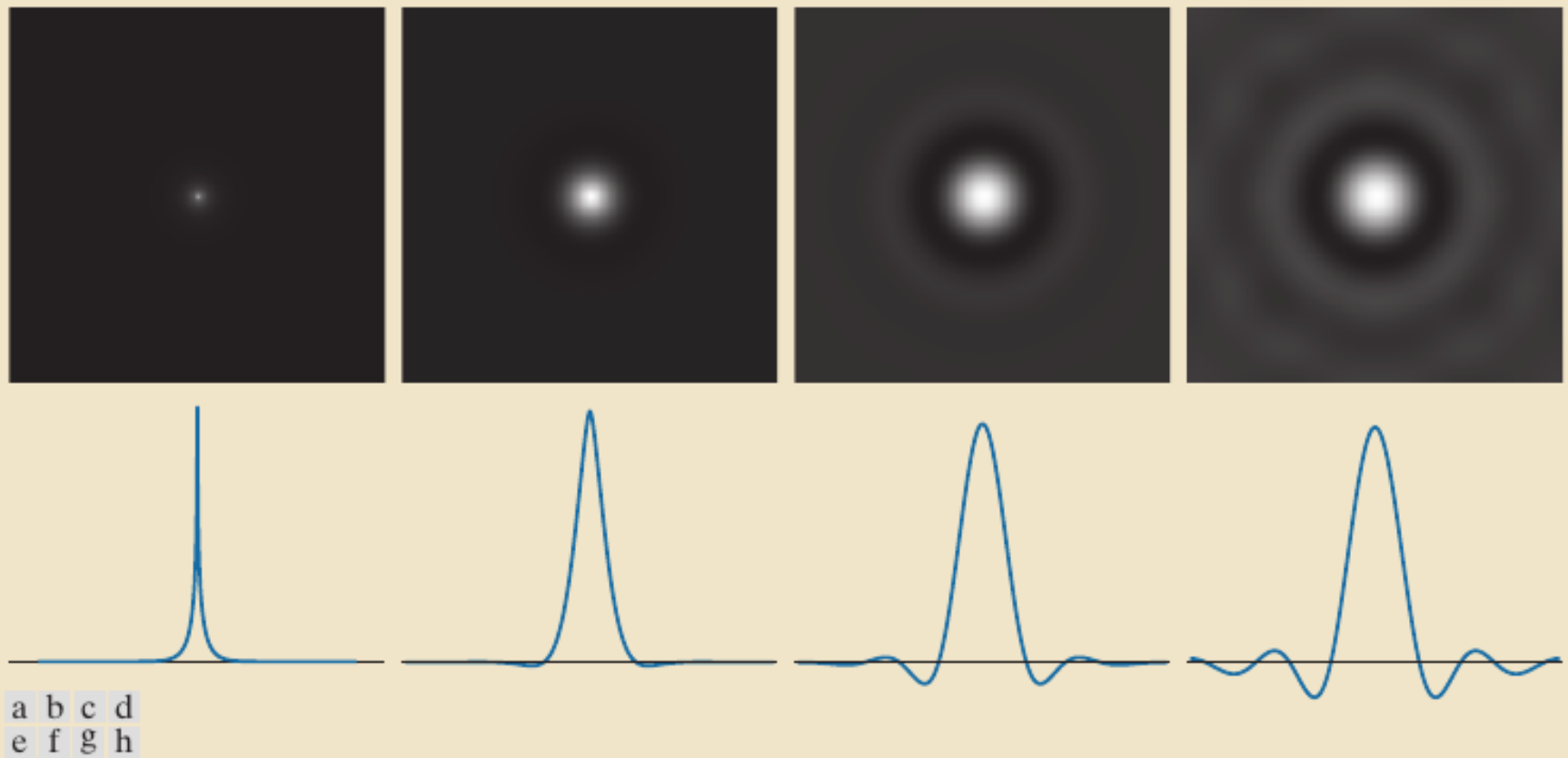


FIGURE 4.47 (a)–(d) Spatial representations (i.e., spatial kernels) corresponding to BLPF transfer functions of size 1000×1000 pixels, cut-off frequency of 5, and order 1, 2, 5, and 20, respectively. (e)–(h) Corresponding intensity profiles through the center of the filter functions.

a b

FIGURE 4.48

(a) Sample text of low resolution (note the broken characters in the magnified view).
(b) Result of filtering with a GLPF, showing that gaps in the broken characters were joined.

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.





a b c

FIGURE 4.49 (a) Original 785×732 image. (b) Result of filtering using a GLPF with $D_0 = 150$. (c) Result of filtering using a GLPF with $D_0 = 130$. Note the reduction in fine skin lines in the magnified sections in (b) and (c).



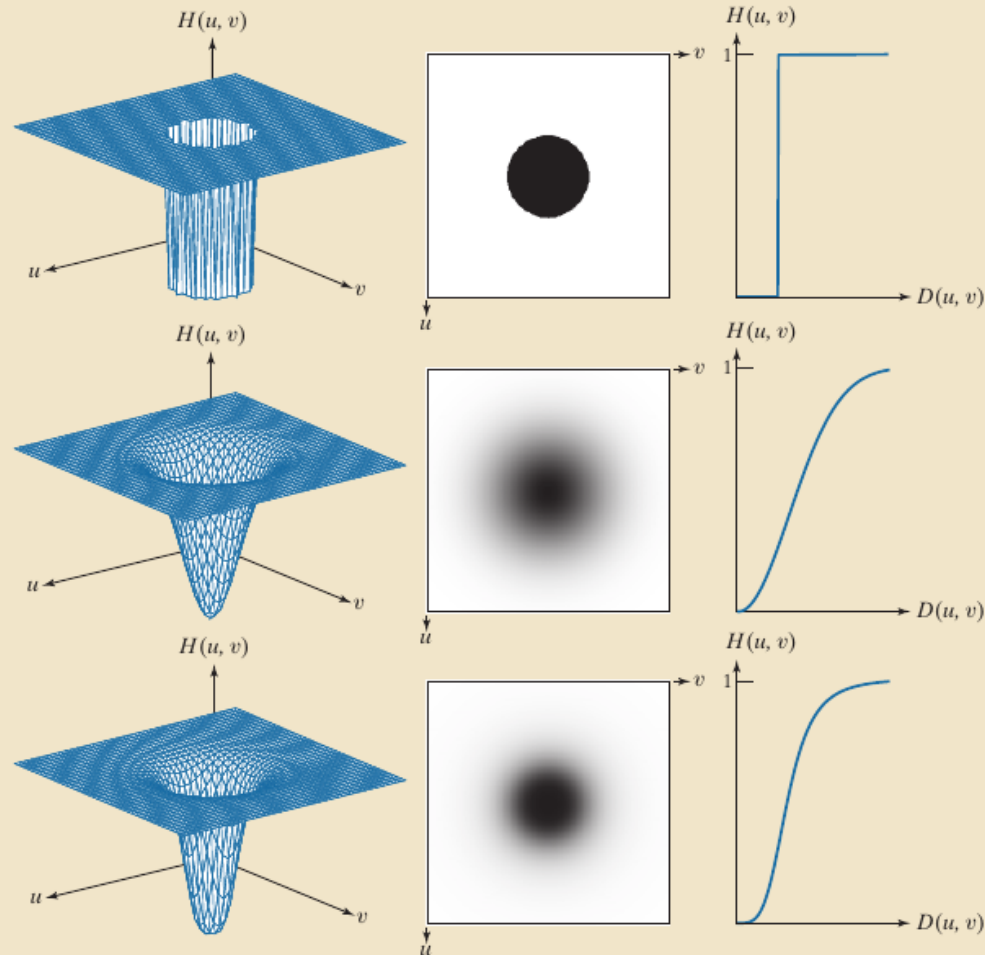
a b c

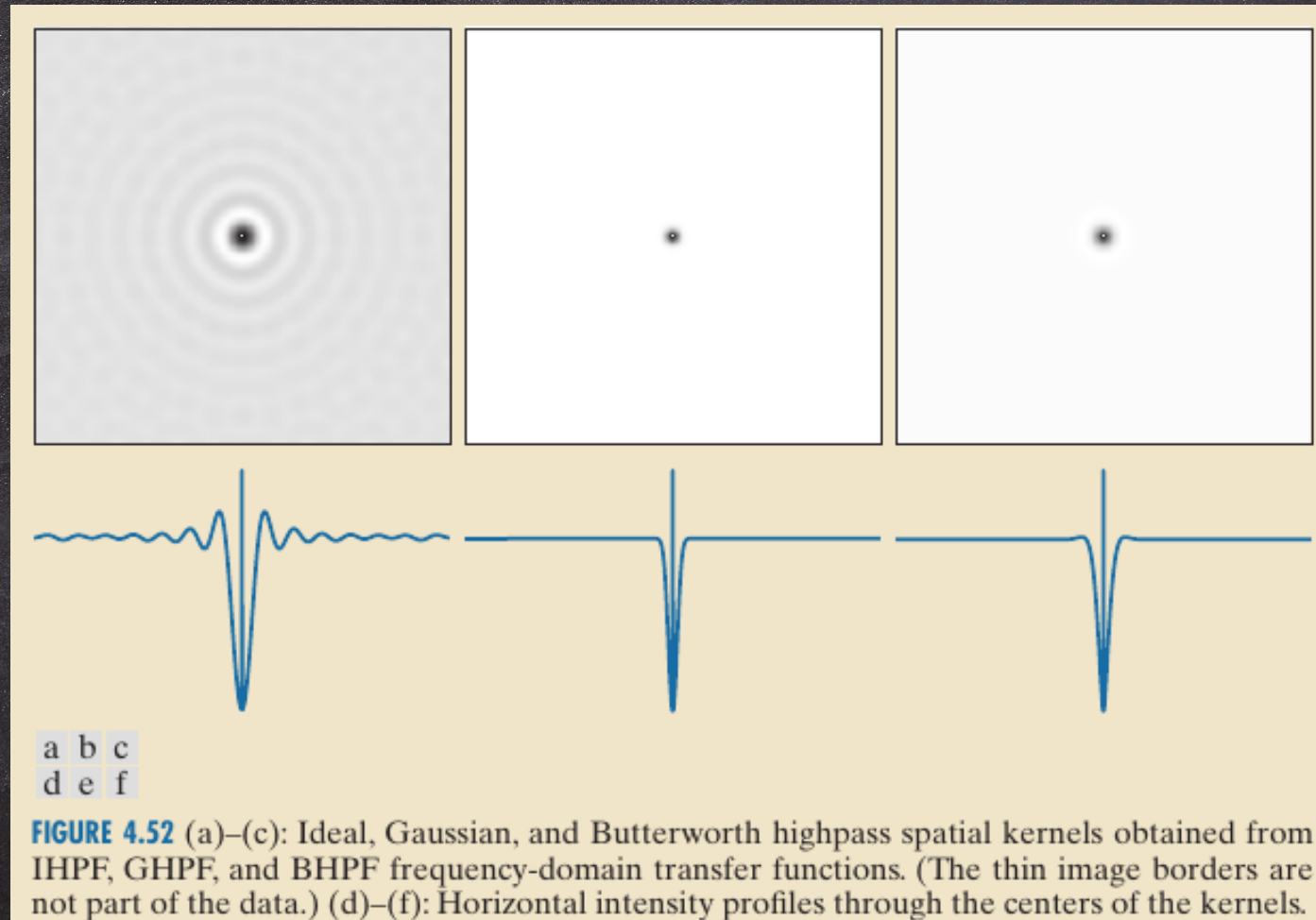
FIGURE 4.50 (a) 808×754 satellite image showing prominent horizontal scan lines. (b) Result of filtering using a GLPF with $D_0 = 50$. (c) Result of using a GLPF with $D_0 = 20$. (Original image courtesy of NOAA.)

a b c
d e f
g h i

FIGURE 4.51

Top row:
Perspective plot,
image, and, radial
cross section of
an IHPF transfer
function. Middle
and bottom
rows: The same
sequence for
GHPF and BHPF
transfer functions.
(The thin image
borders were
added for clarity.
They are not part
of the data.)





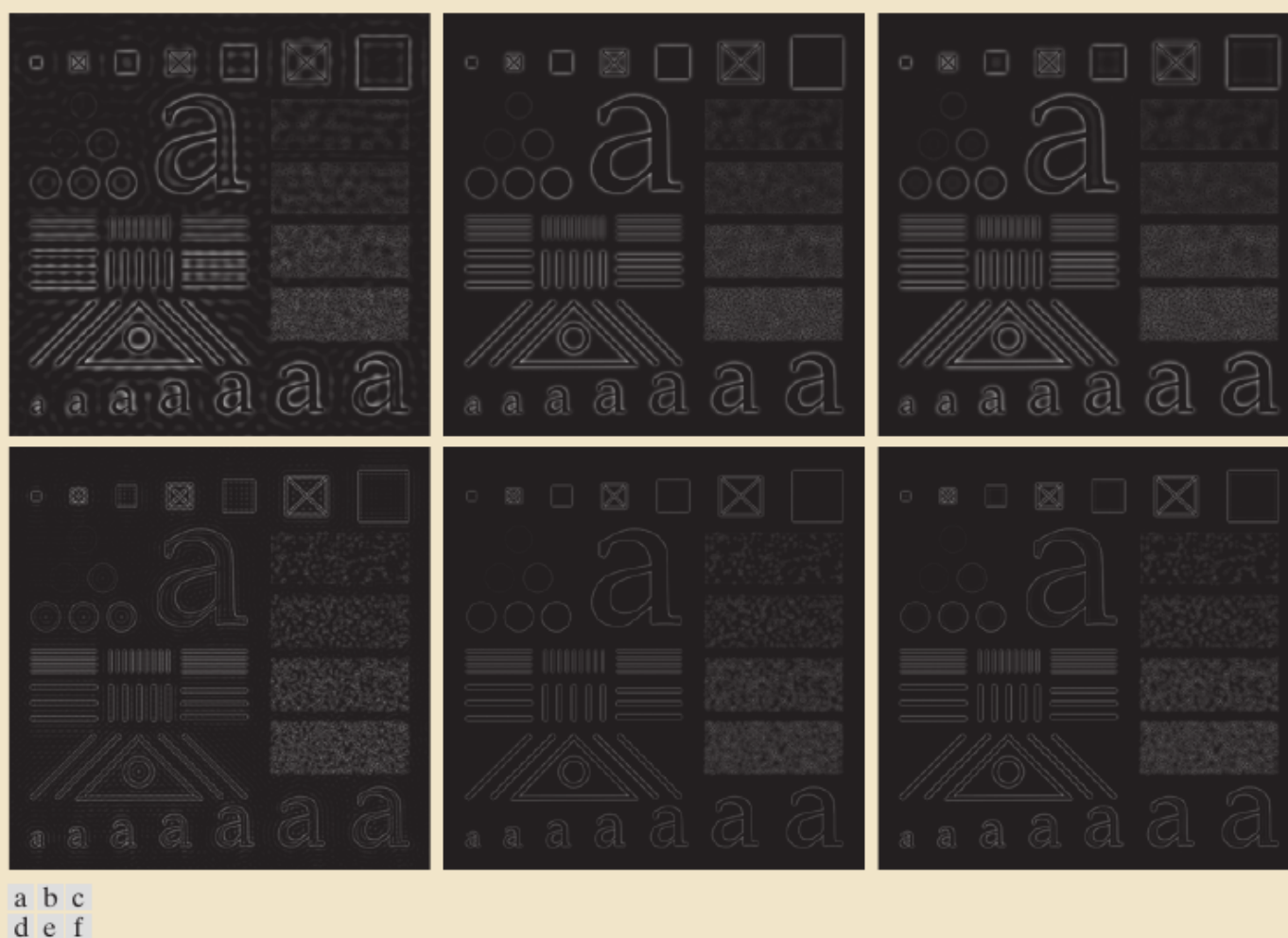


FIGURE 4.53 Top row: The image from Fig. 4.40(a) filtered with IHPF, GHPF, and BHPF transfer functions using $D_0 = 60$ in all cases ($n = 2$ for the BHPF). Second row: Same sequence, but using $D_0 = 160$.

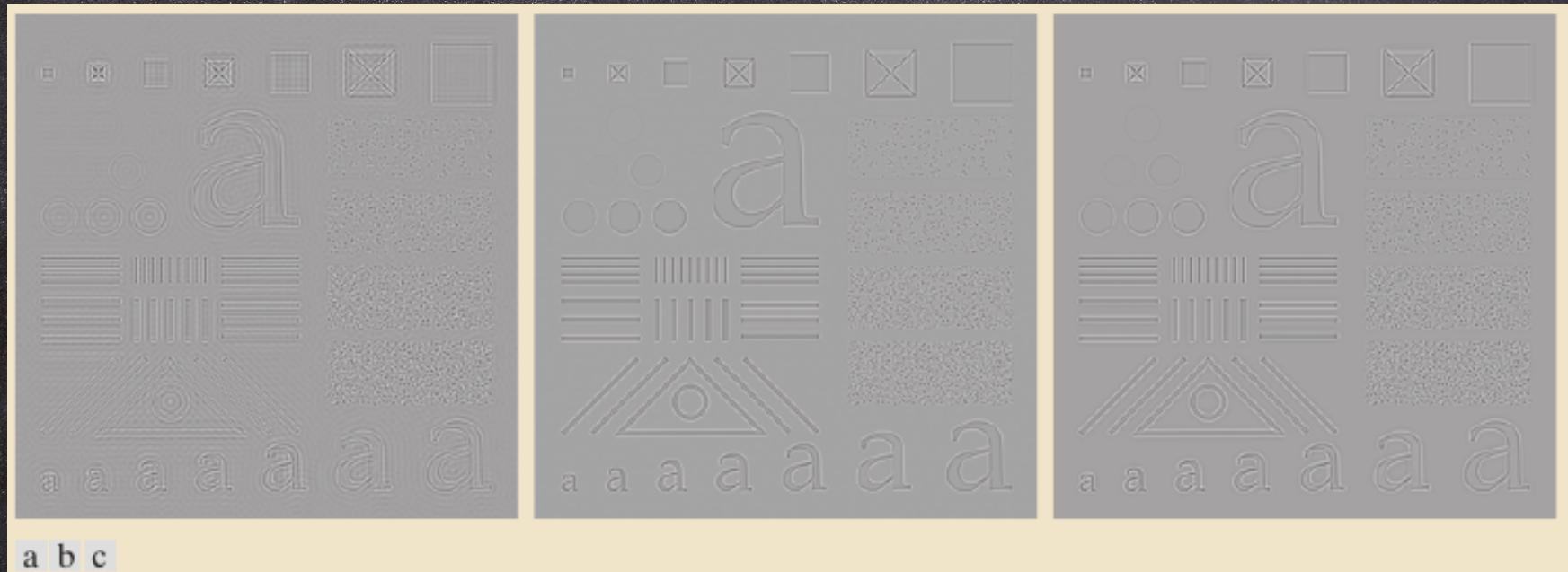


FIGURE 4.54 The images from the second row of Fig. 4.53 scaled using Eqs. (2-31) and (2-32) to show both positive and negative values.



a b c

FIGURE 4.55 (a) Smudged thumbprint. (b) Result of highpass filtering (a). (c) Result of thresholding (b). (Original image courtesy of the U.S. National Institute of Standards and Technology.)

$$\begin{aligned}g(x, y) &= f(x, y) + c\nabla^2 f(x, y) \\&= \mathfrak{F}^{-1}((1 + 4\pi^2 D^2(u, v))F(u, v))\end{aligned}$$

a b

FIGURE 4.56

(a) Original, blurry image.

(b) Image enhanced using the Laplacian in the frequency domain.

Compare with Fig. 3.46(d). (Original image courtesy of NASA.)



$$g(x, y) = \mathfrak{F}^{-1}((k_1 + k_2 H_{\text{HP}}(u, v))F(u, v))$$

a b
c d

FIGURE 4.57

(a) A chest X-ray.
(b) Result of filtering with a GHPF function.
(c) Result of high-frequency-emphasis filtering using the same GHPF. (d) Result of performing histogram equalization on (c). (Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

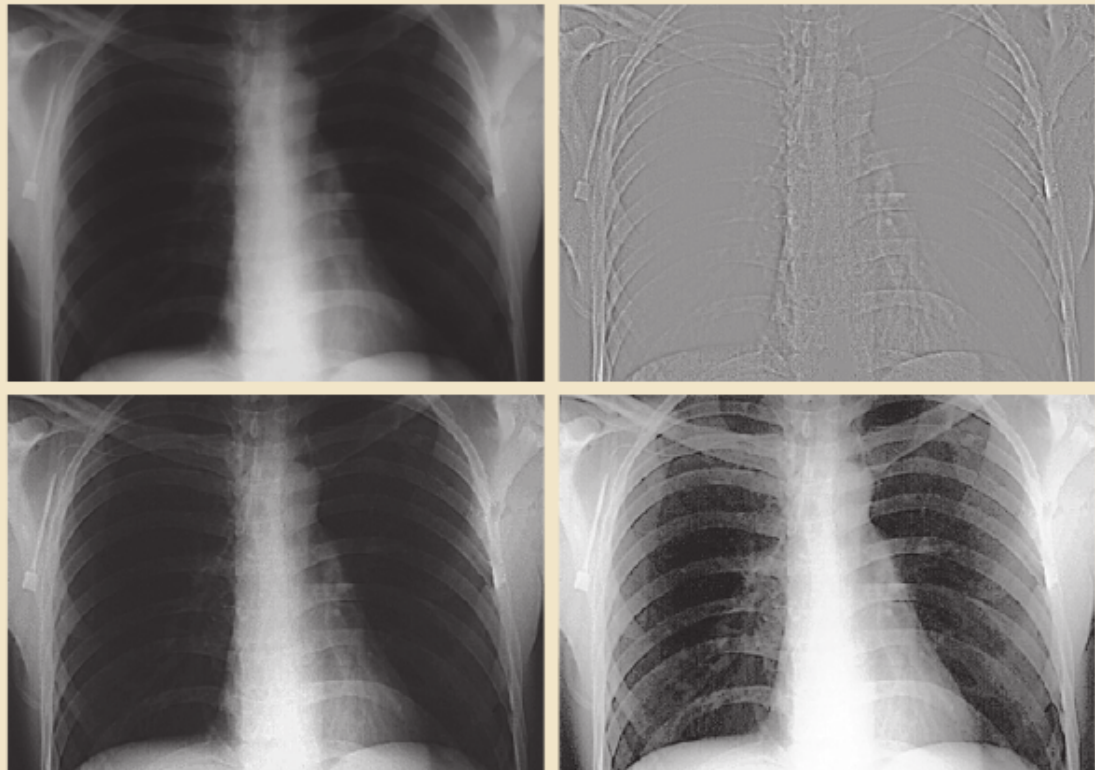
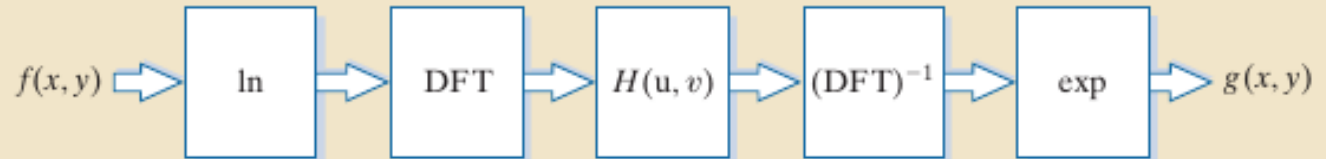


FIGURE 4.58

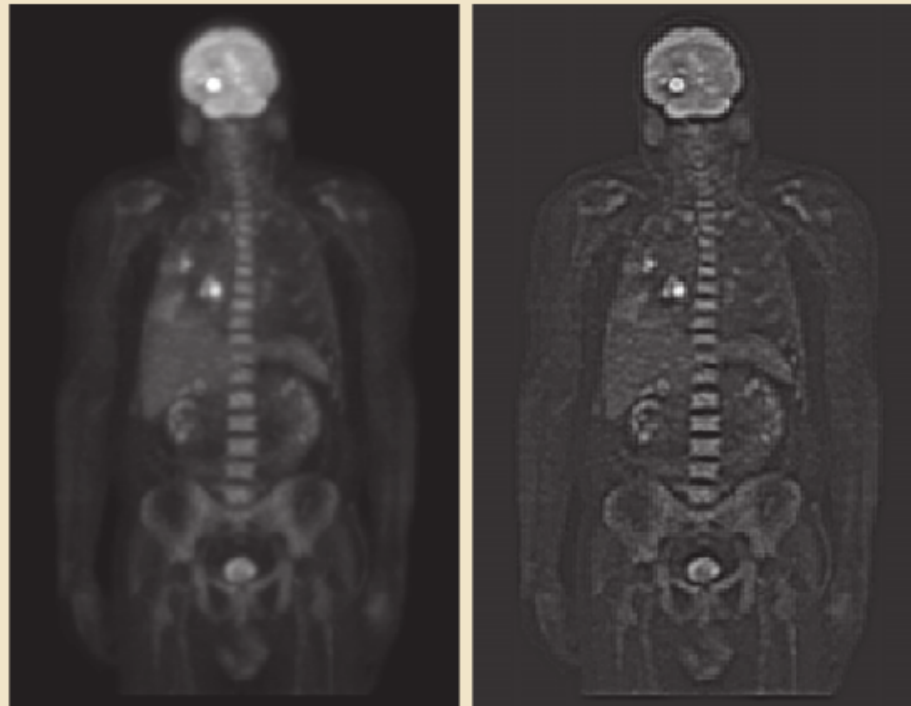
Summary of steps in homomorphic filtering.



a b

FIGURE 4.60

(a) Full body PET scan. (b) Image enhanced using homomorphic filtering. (Original image courtesy of Dr. Michael E. Casey, CTI Pet Systems.)



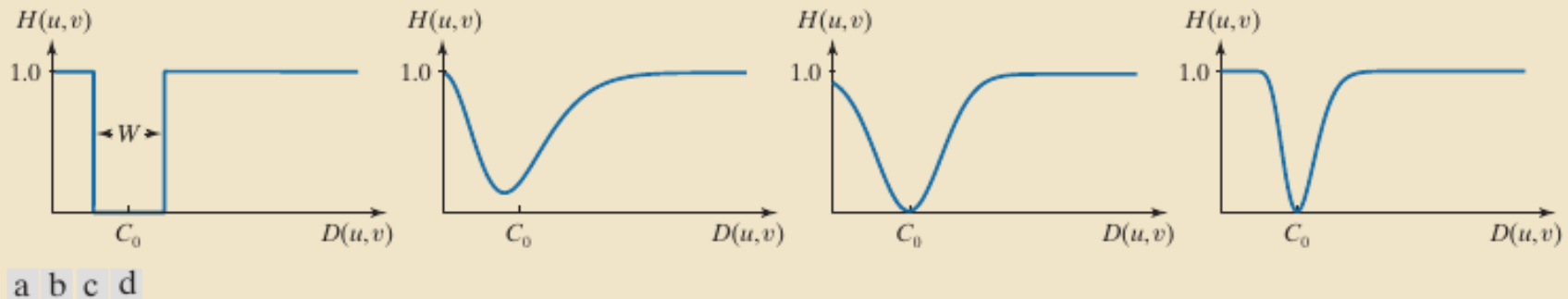


FIGURE 4.61 Radial cross sections. (a) Ideal bandreject filter transfer function. (b) Bandreject transfer function formed by the sum of Gaussian lowpass and highpass filter functions. (The minimum is not 0 and does not align with C_0 .) (c) Radial plot of Eq. (4-149). (The minimum is 0 and is properly aligned with C_0 , but the value at the origin is not 1.) (d) Radial plot of Eq. (4-150); this Gaussian-shape plot meets all the requirements of a bandreject filter transfer function.

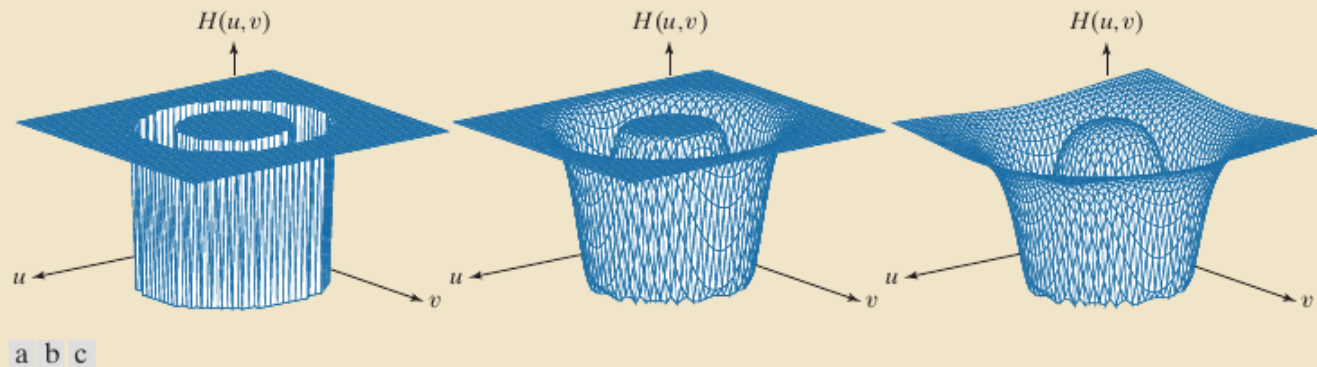
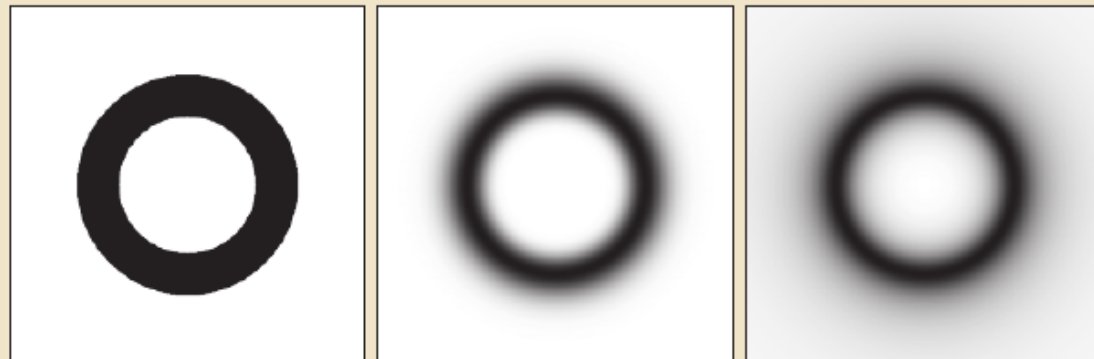


FIGURE 4.62 Perspective plots of (a) ideal, (b) modified Gaussian, and (c) modified Butterworth (of order 1) band-reject filter transfer functions from Table 4.7. All transfer functions are of size 512×512 elements, with $C_0 = 128$ and $W = 60$.

a b c

FIGURE 4.63

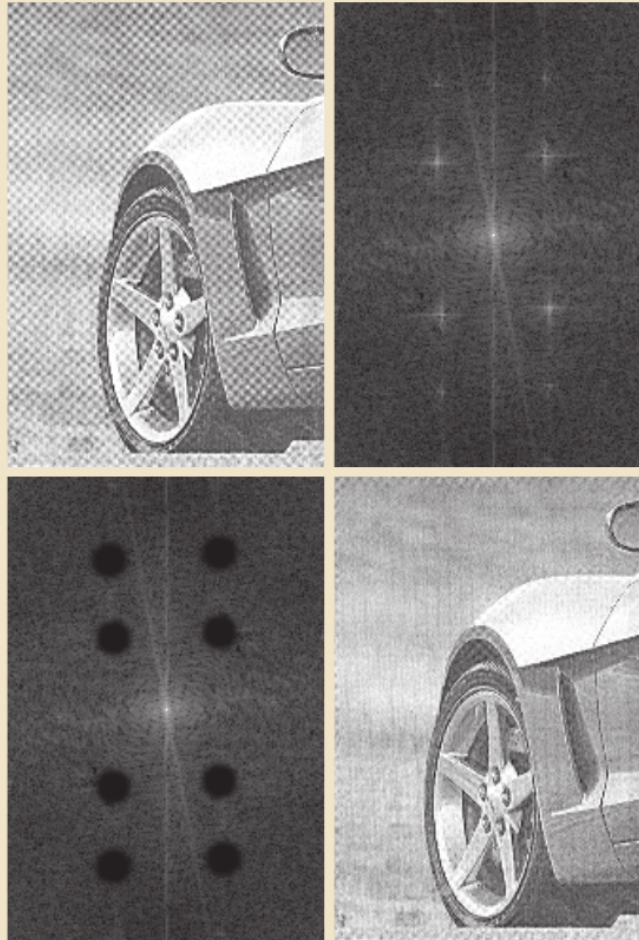
(a) The ideal, (b) Gaussian, and (c) Butterworth bandpass transfer functions from Fig. 4.62, shown as images. (The thin border lines are not part of the image data.)



a b
c d

FIGURE 4.64

(a) Sampled newspaper image showing a moiré pattern.
(b) Spectrum.
(c) Fourier transform multiplied by a Butterworth notch reject filter transfer function.
(d) Filtered image.



a b
c d

FIGURE 4.65

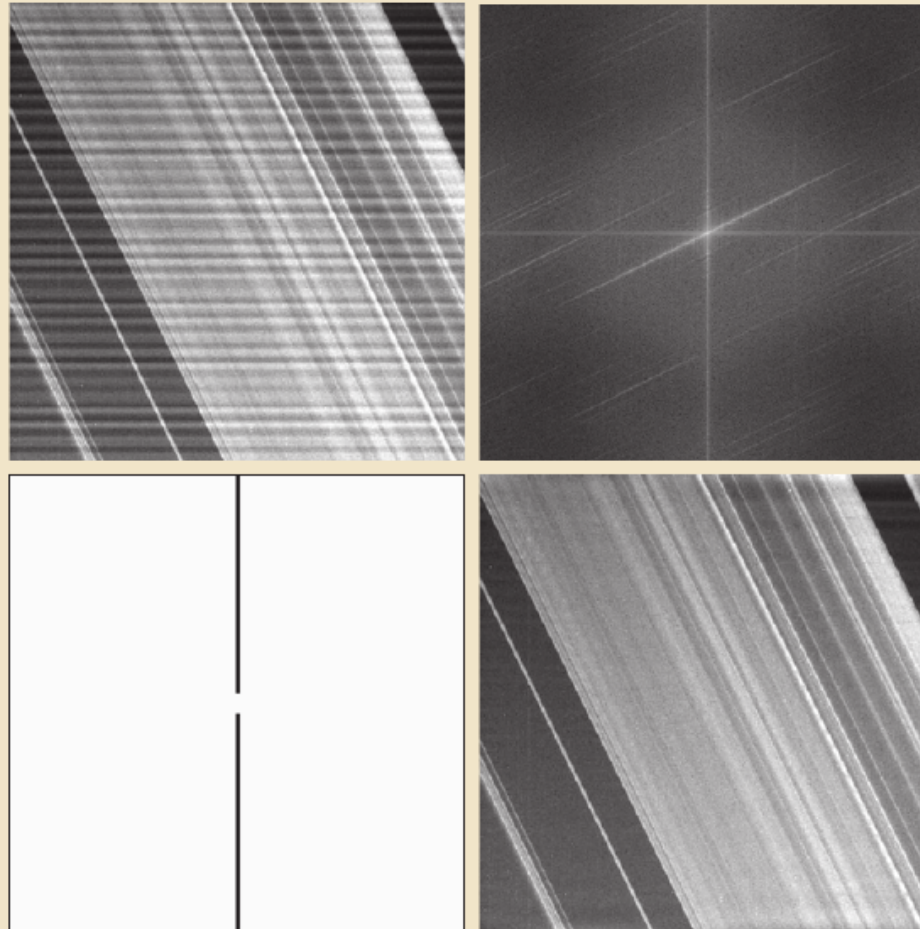
(a) Image of Saturn rings showing nearly periodic interference.

(b) Spectrum. (The bursts of energy in the vertical axis near the origin correspond to the interference pattern).

(c) A vertical notch reject filter transfer function.

(d) Result of filtering.

(The thin black border in (c) is not part of the data.) (Original image courtesy of Dr. Robert A. West, NASA/JPL.)



a b

FIGURE 4.66

(a) Notch pass filter function used to isolate the vertical axis of the DFT of Fig. 4.65(a).

(b) Spatial pattern obtained by computing the IDFT of (a).

